

Probability, Statistics & Data Analysis

Lecture 6: Introduction to NHST and z-tests

**The combination of some data and
an aching desire for an answer does
not ensure that a reasonable answer
can be extracted from a given body
of data**

John Tukey

Lecture 6: Introduction to NHST and z-tests

What we will cover

- Null hypothesis statistical testing
 - Tailedness
 - NHST outcomes
 - Probability of Type I and Type II errors
 - Drawing proper conclusions from NHST
- Using confidence intervals
- One sample z-test
- Independent two-sample z-test
- Dependent two-sample z-test





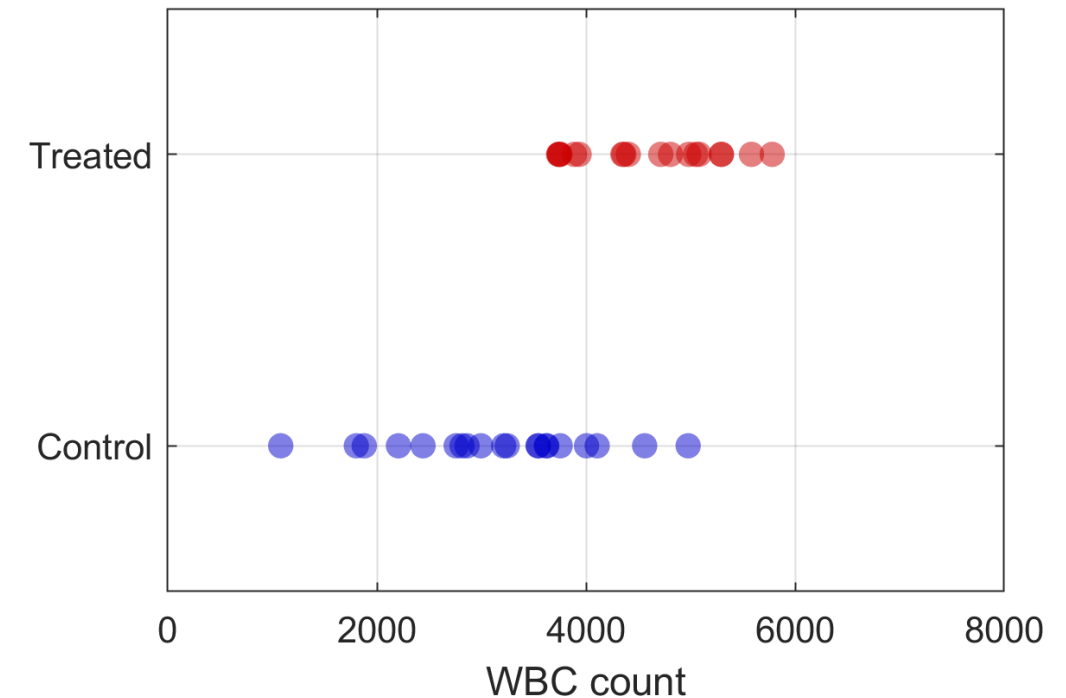
Using confidence intervals

To determine 'statistical differences'

Motivation - WBC count



- A side-effect of a chemotherapy drug is a reduction in white blood cell (WBC) count
- A new drug is proposed (a colony stimulating factor) that promises to reduce this side-effect
- Clinical trial
 - Test some patients without the medicine
 - Test some patients with the medicine
 - Quantify number of WBC from blood tests
- ‘How well does it work’?



“How well does it work?”

- This question is a core motivation for a lot of statistical analysis in bioengineering and indeed much of science
- For example, we seek answers to
 - Does the drug reduce side effects?
 - Is it as effective as the drug it might replace?
- These answers, along with information on cost and availability would allow healthcare providers and clinicians to decide whether to use the new drug or stick with the old one
- Perhaps surprisingly, a large amount statistical analysis is focussed on the simpler question

“Does the drug produce a different outcome, on average”

- We actually already kind of know how to do this....

WBC Count Confidence Interval

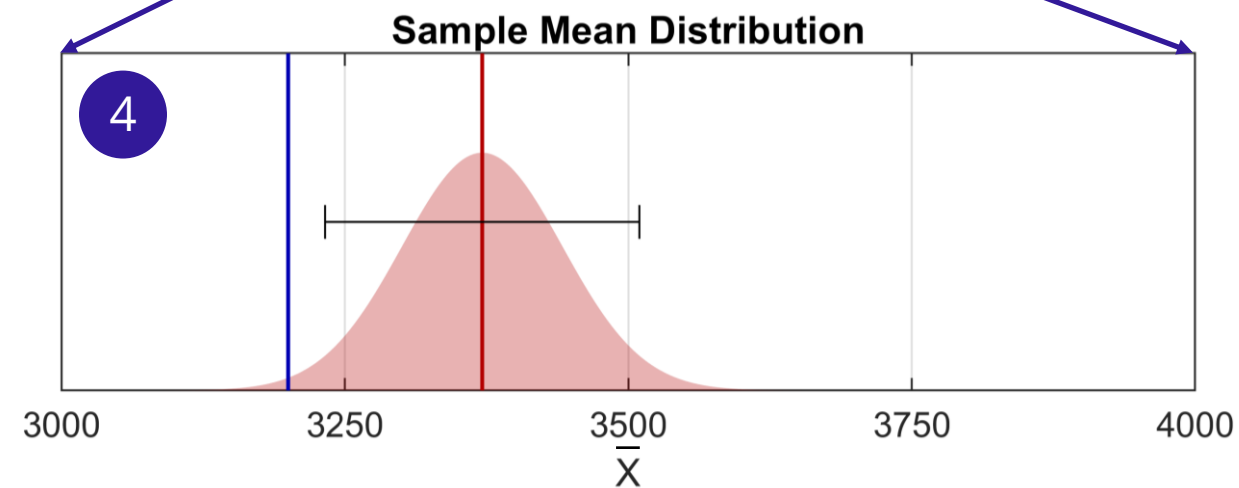
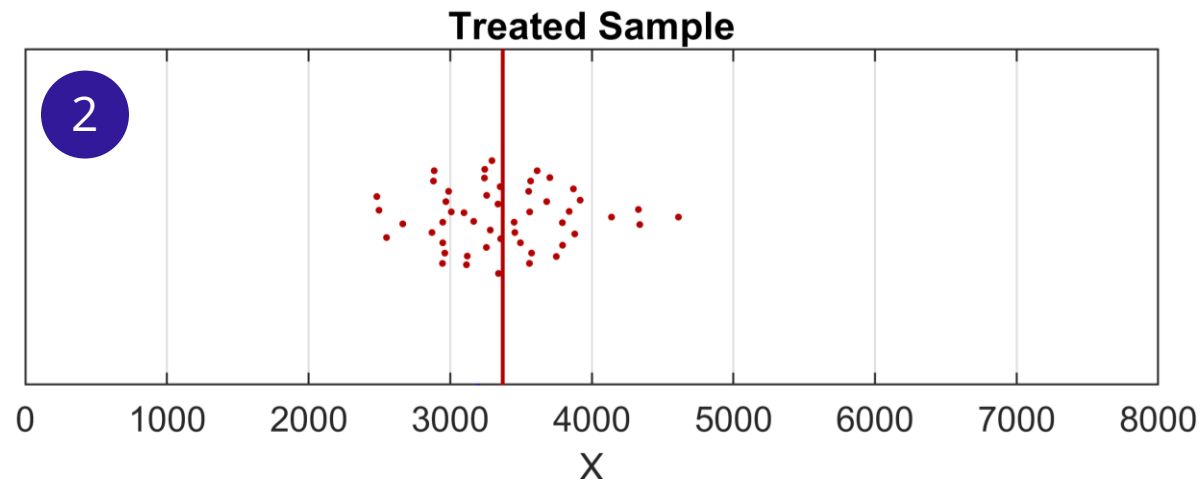
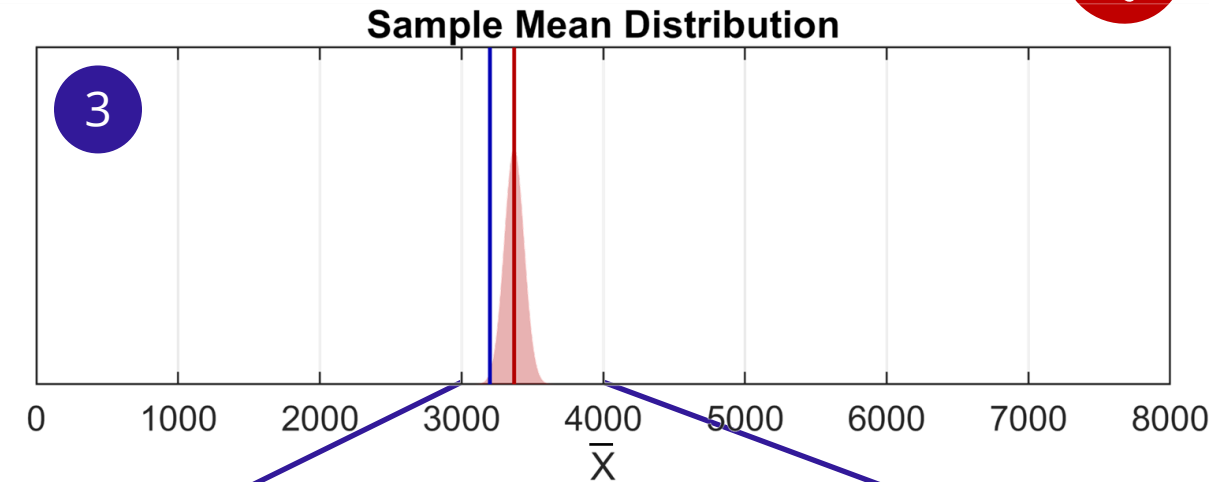
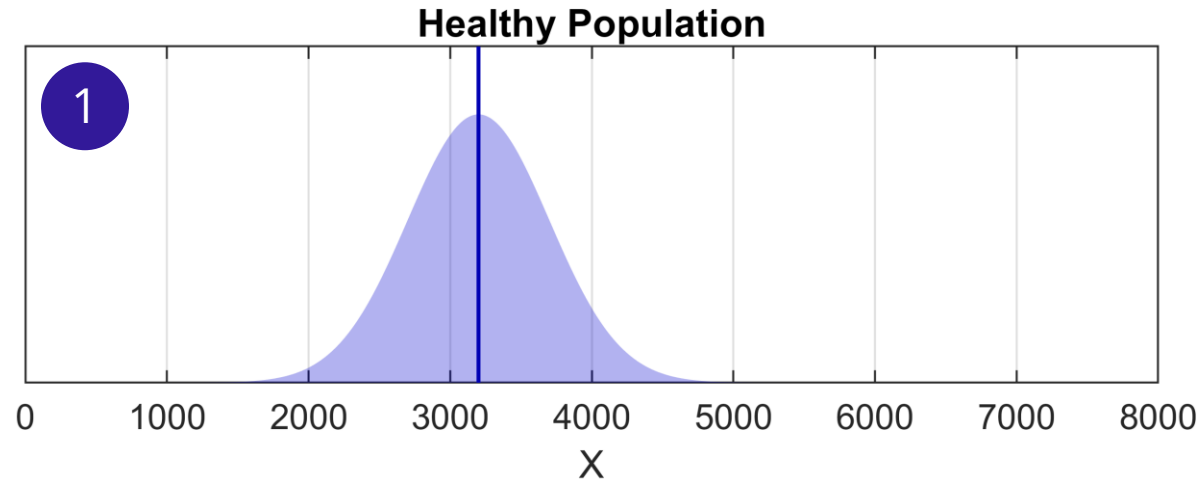


- X = number of white blood cells
- Assume that for untreated individuals, X_0 , we have a sufficiently large dataset over the past 30 years that we know the population distribution ($\mu_0 = 3,200$, $\sigma_0 = 500$)
- Clinical trial yields $n = 50$ values of X_1 for treated individuals with $\bar{x}_1 = 3,371$
- Assume that the treatment doesn't affect the SD, hence $\sigma_1 = \sigma_0 = 500$

Using $\alpha = 0.05$ (95% confidence interval),
is it likely that we observed this value of
 $\bar{x}_1$ if $\mu_1 = \mu_0$?

- μ_0 is not in the 95% confidence interval on $\bar{x}_1$
- If $\mu_1 = \mu_0$, we would only expect this to occur 5 in 100 times
- Let's visualise

WBC Count Visualisation



- If $\mu_1 = \mu_0$, we would only expect the *CI* on $\bar{x}_1$ not to contain μ_0 5 in 100 times
- We conclude that “ μ_1 is statistically different to μ_0 ”

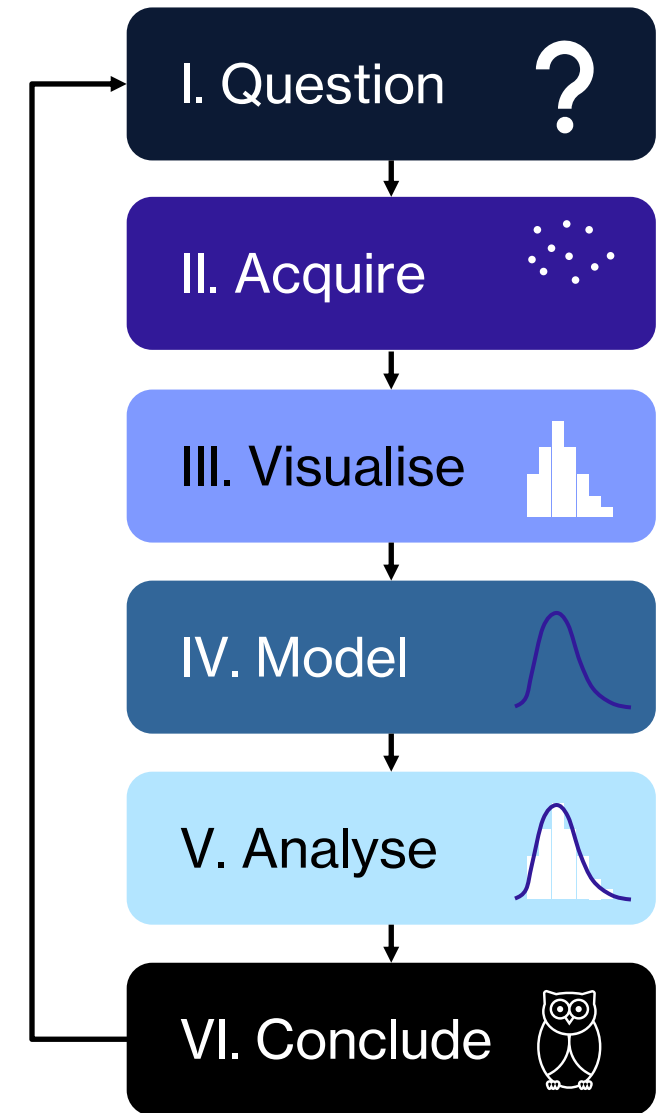


Statistical Null-Hypothesis Significance Testing

NHST

The Null-Hypothesis Approach

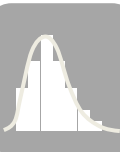
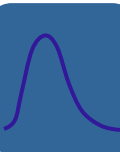
1. Rather than asking 'How well does it work'? we ask 'does it work better than no drug'
 - Note this has a binary outcome, yes or no
- Generate a null hypothesis, H_0
 - H_0 : there is no difference between the treated and untreated patients (hence null)
- Compare to an alternative hypothesis, H_1
 - H_1 : there is a difference between the treated and untreated patients
2. Gather a sample
3. Visualise it
4. Choose a model
5. Use the model to **estimate whether it is likely that we got the sample we did, if H_0 is true.** Hence 'Reject' or 'do not reject' H_0 .
6. Draw some conclusions



Null-Hypothesis Models



- For the alternative hypothesis ' H_1 ' we must define a model
- Initially, for all models we assume that the data are sampled from a population with a normal distribution
 - We'll look at how to check if this is valid and what to do if not in future lectures
- Generally, 3 options $\neq$, $>$, $<$
- Examples using WBC count
 - 'The drug has an effect'
 - The model H_1 would be: $\mu_1 \neq \mu_0$
 - 'The drug increases WBC count'
 - The model H_1 would be: $\mu_1 > \mu_0$
 - 'The drug decreases WBC count'
 - The model H_1 would be: $\mu_1 < \mu_0$
- More on this on Slide 20



Null-Hypothesis Analysis

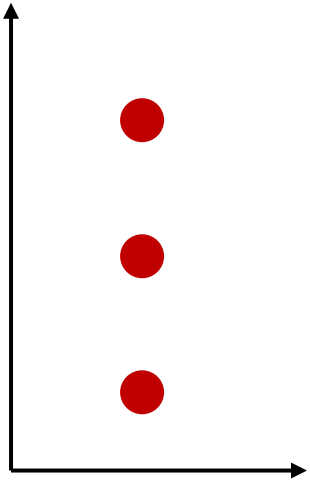
- Having gathered the sample, defined the null and alternative hypotheses, chosen a model and chosen α ...
1. We calculate a **test statistic**

E.g. Difference between mean and zero

$$\text{Test statistic} = \frac{\text{Effect}}{\text{Error}} = \frac{\text{Signal}}{\text{Noise}}$$

E.g. The standard error on the mean as an estimate of our error

2. Then either
 - a) Evaluate whether the test statistic is in a 'rejection region'
 - b) Estimate a probability of getting the observed value of the test statistic if the null hypothesis is true



The one sample Z-test

Analysing one sample

WBC Null-Hypothesis Approach



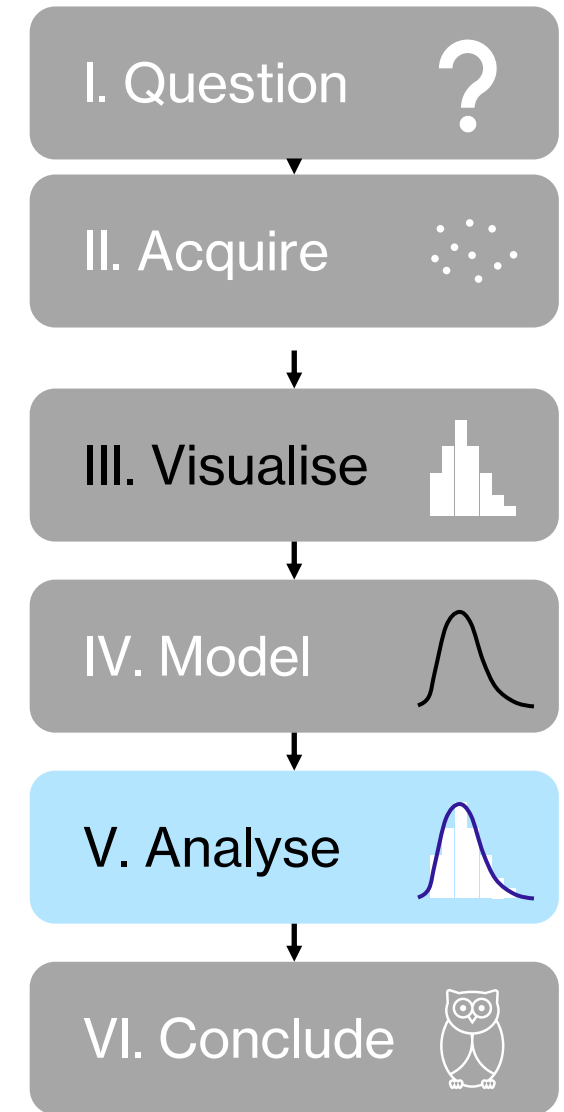
- ‘Does the drug affect WBC count?’
 - H_0 : there is no difference between the population means of the treated and untreated populations
$$\mu_0 = \mu_1$$
 - H_1 : there is a difference between the population means of the treated and untreated populations
$$\mu_0 \neq \mu_1$$
- Model: $\bar{X}_1 \sim N(\mu_1, \sigma_1^2/n)$
- The test statistic for a z-test is the Z-statistic

$$Z = \frac{\bar{X} - \mu_0}{\sigma_{\bar{X}}}$$

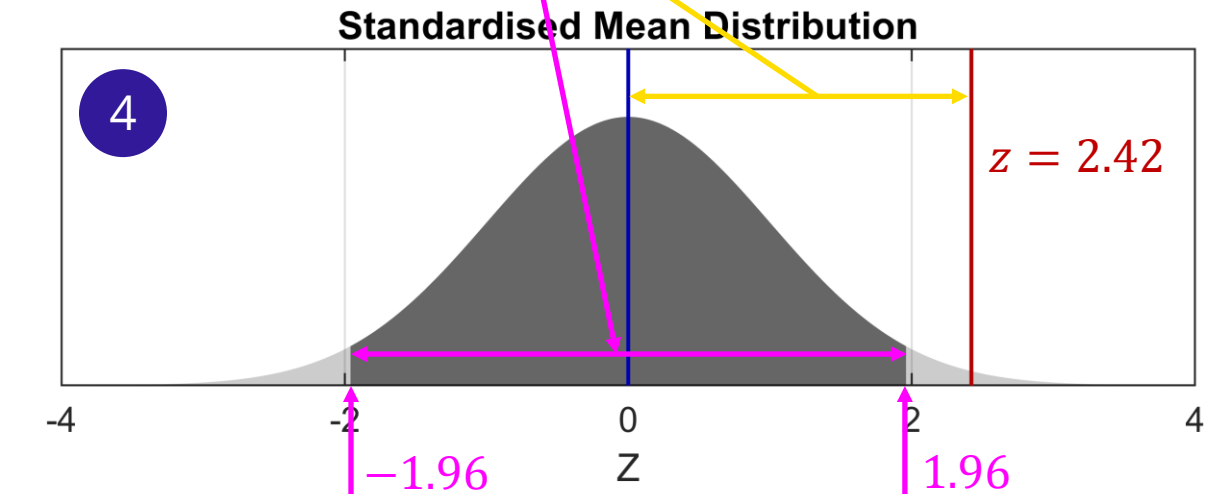
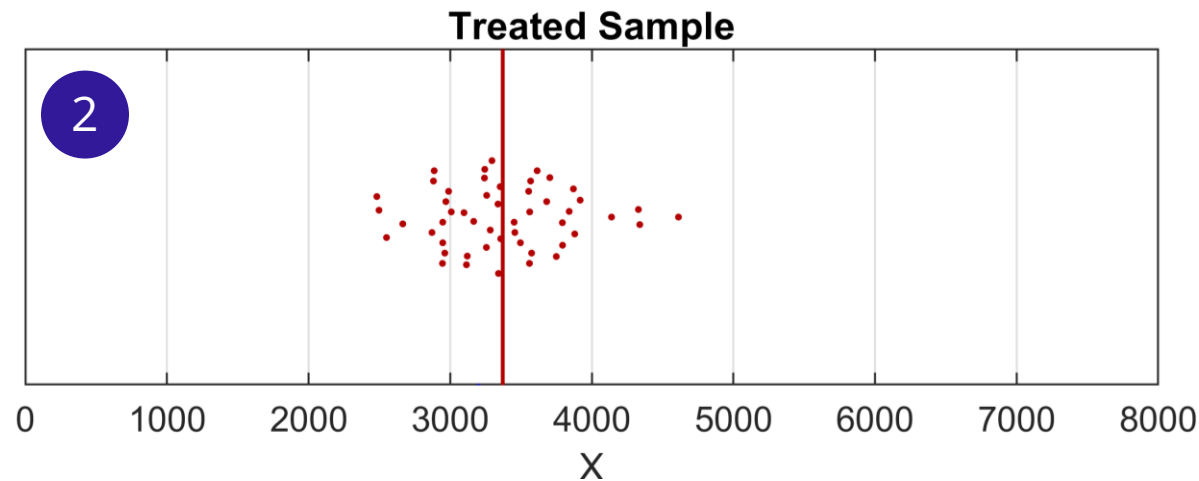
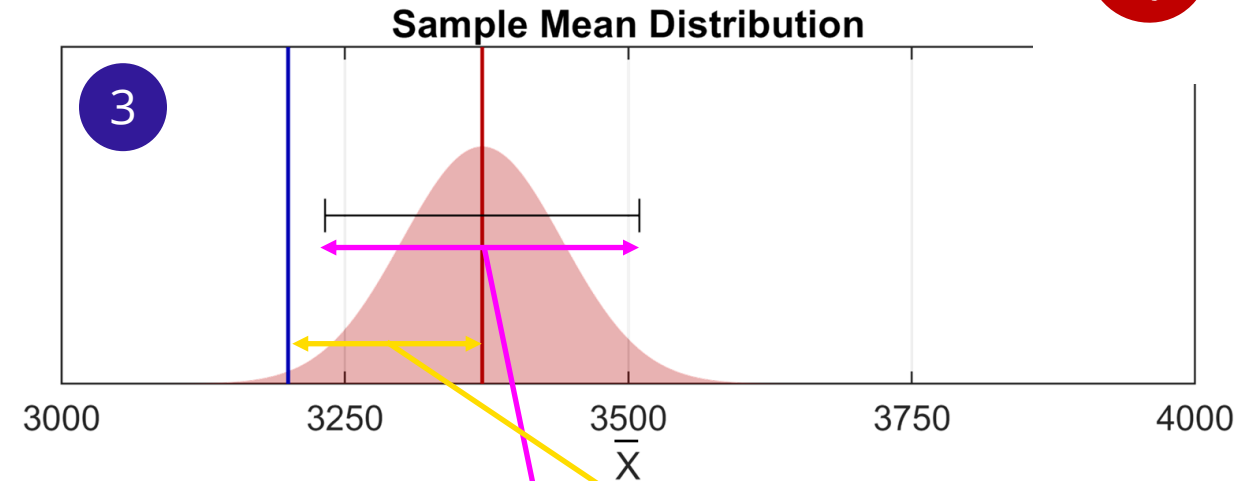
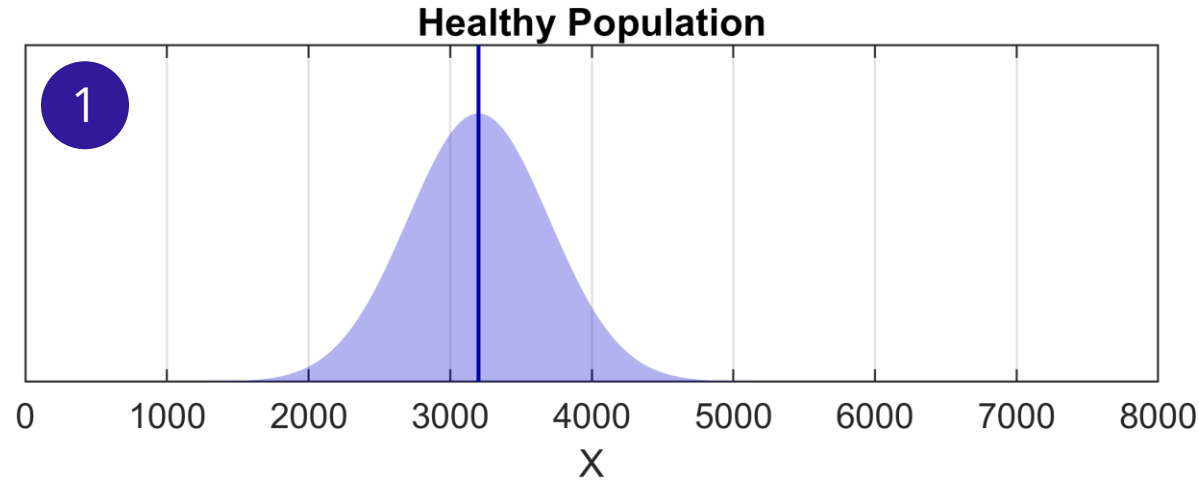
μ_0 is a known
reference value

- Or specifically for this case

$$z = \frac{\bar{x}_1 - \mu_0}{\sigma_{\bar{x}_1}} = \frac{3,371 - 3,200}{500/\sqrt{50}} = 2.42$$



WBC Count Z-test Visualisation



- $1.96 = z_{1-\alpha/2}$ our 'critical z-value' determined by choice of $\alpha = 0.05$
- **z-statistic** is determined by the data (difference between means, normalised)



WBC Count Z-test with rejection region

- Define a 'rejection region'

$$z > z_{1-\frac{\alpha}{2}} \text{ and } z < -z_{1-\frac{\alpha}{2}}$$

$$z_{1-\frac{\alpha}{2}} = z_{crit} = \Phi^{-1}\left(1 - \frac{\alpha}{2}\right)$$

- Or more neatly

$$|z| > z_{1-\frac{\alpha}{2}} \text{ or } |z| > z_{crit}$$

- If our z-statistic is in the rejection region, we

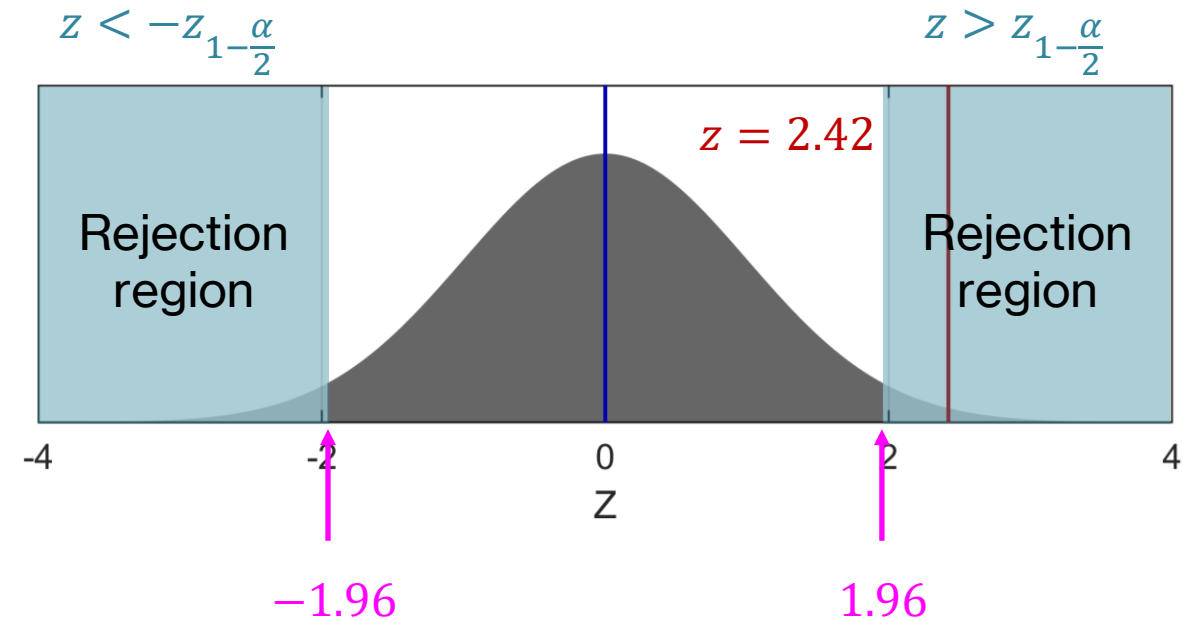
“reject the null hypothesis”

and hence conclude that

“ μ_1 is statistically different to μ_0 ”

- If our z-statistic is not in the rejection region, we

“do not reject the null hypothesis”



WBC Count Z-test with 'p-value'



- If the null hypothesis is true

$$\mu_0 = \mu_1$$

- What was the probability (before we did the experiment) that we observed this outcome or one that is more extreme?
 - Extreme means a larger $|z|$

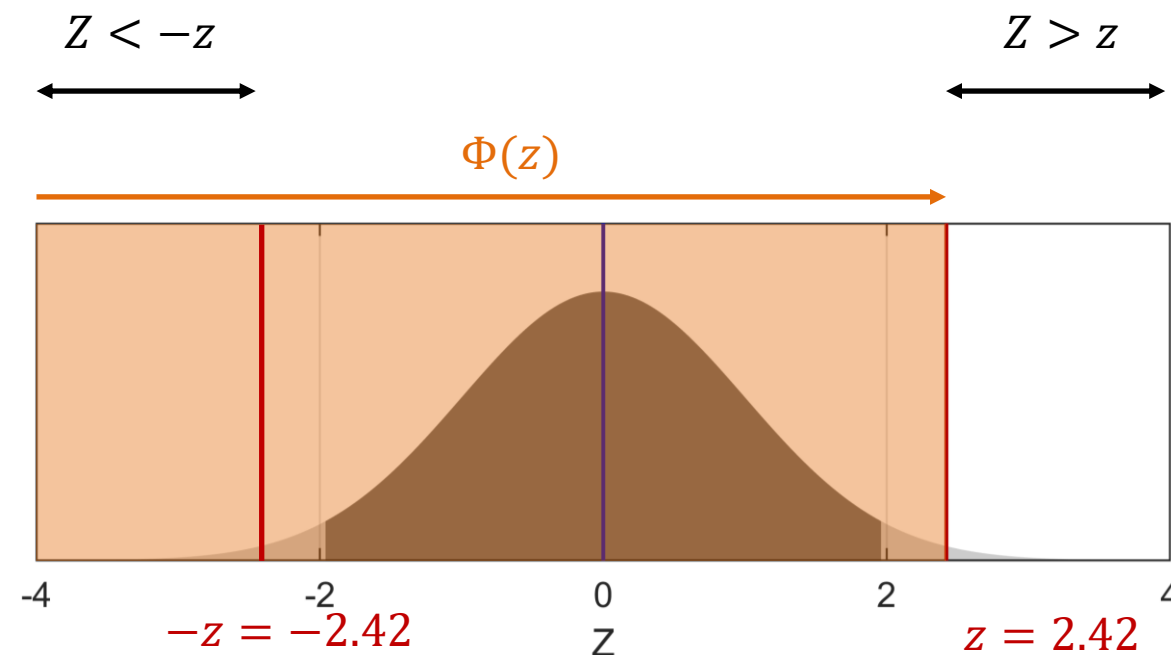
- $z = 2.42$

$$p = P(|Z| > |z|) = P(Z > z) + P(Z < -z)$$

$$p = 2P(Z > |z|)$$

$$p = 2(1 - \Phi(|z|))$$

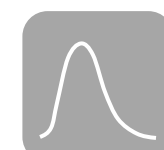
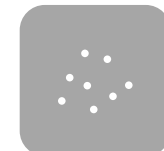
- $\Phi(2.42) = 0.9922$, $p = 0.0156$
- For equivalence to CI, compare to α
- As $p < \alpha$, we “*reject H_0* ”
- If $p > \alpha$, we “*do not reject H_0* ”



Notes on interpretation

In the example, we “*rejected the null hypothesis with $\alpha = 0.05$* ”

1. The value α is commonly termed the ‘significance level’
 - I don’t like this terminology, see next page
2. This means that 1 in 20 times, we would expect to get the wrong answer (reject H_0 when it is true)
3. If we had “*not rejected H_0* ”, it would not mean that there **is** no difference
 - It only means that the effect we observed cannot be determined to be different (or that we did not detect a difference)
 - We do not say “*we rejected H_1* ”
 - Note that technically μ is always different to $\bar{X}$!
4. It is typically also important to consider the ‘spread’ of the data
 - Does the drug have a more variable response, leading to some people having worse outcomes?
5. A *statistical difference* in means is not the same as *practically significant*
 - E.g. is the ~5% average increase we observed actually beneficial?

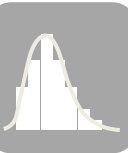
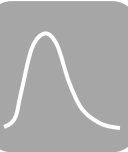
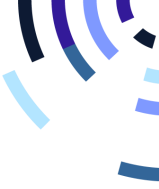


Statistically Significant vs Different

- The term ‘statistically significant’ is widely used when we reject the null hypothesis
- However, this leads to misinterpretation due to the difference between **statistical difference** and **practical significance**
- Instead, I **insist** you use the term

‘Statistically different’

- **Statistical difference** applies only to the outcome of the test
- **Practical significance** is a more subjective outcome, that should and must be considered separately
- If someone tells you that you should use ‘statistically significant’, carry out a simple NHST:
 - H_0 : their terminology is better
 - Reject H_0





Tailedness

Directionality of our hypothesis

2-tailed test

- In the previous section, we carried out a 2-tailed test
- H_0 : 'The drug does **not change** WBC count'

$$\mu_0 = \mu_1$$

- H_1 : 'The drug **changes** WBC count'

$$\mu_1 \neq \mu_0$$

- We had to calculate two probabilities:

$$P(Z > z) \text{ and } P(Z < -z)$$

- There are two 'tails' of the distribution

- Area of the two tails equals α .

$$|z| \geq z_{1-\frac{\alpha}{2}}$$

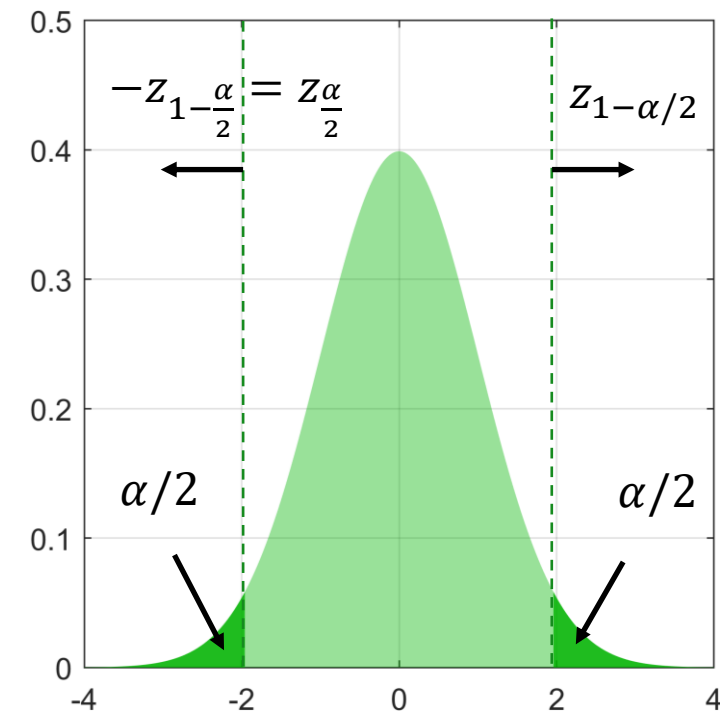
Rejection region

$$p = 2(1 - \Phi(z))$$

P-value



$$z_{1-\frac{\alpha}{2}} = z_{crit}$$



1-tailed tests: Upper tailed

- If we only wanted to know
- H_0 : 'The drug does **not increase** WBC count'

$$\mu_1 \leq \mu_0$$

- H_1 : The drug **increases** WBC count'

$$\mu_1 > \mu_0$$

- We only need to calculate the probability $P(Z > z)$
- There is one 'tail' of the distribution
- Area of the tails equals α , hence

$$z \geq z_{1-\alpha}$$

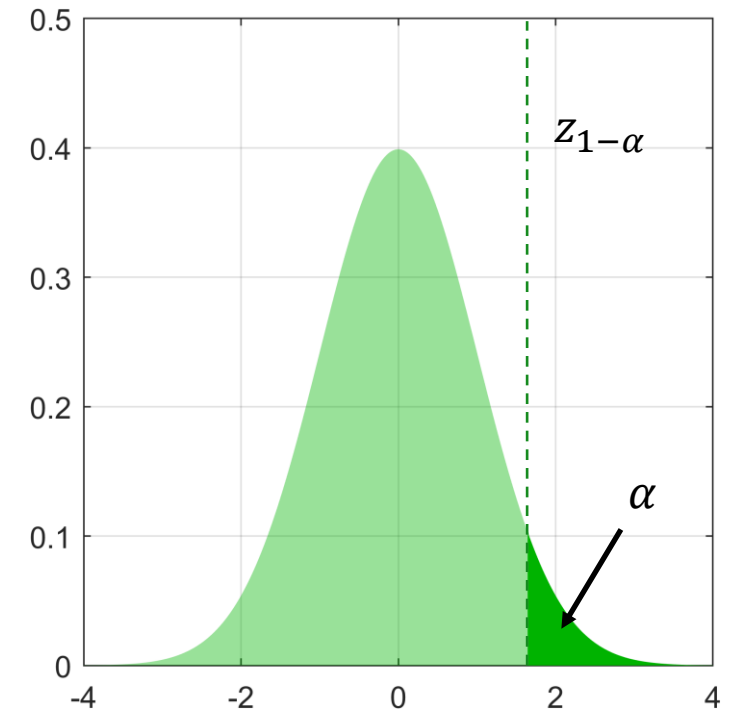
Rejection region

$$p = 1 - \Phi(z)$$

P-value



$$z_{1-\alpha} = z_{crit}$$



1-tailed tests: Lower tailed

- If we only wanted to know
- H_0 : 'The drug does **not decrease** WBC count'

$$\mu_1 \geq \mu_0$$
- H_1 : The drug **decreases** WBC count'

$$\mu_1 < \mu_0$$
- We only need had to calculate the probability $P(Z < -z)$
- There is one 'tail' of the distribution
- Area of the tails equals α , hence

$$z \leq z_\alpha$$

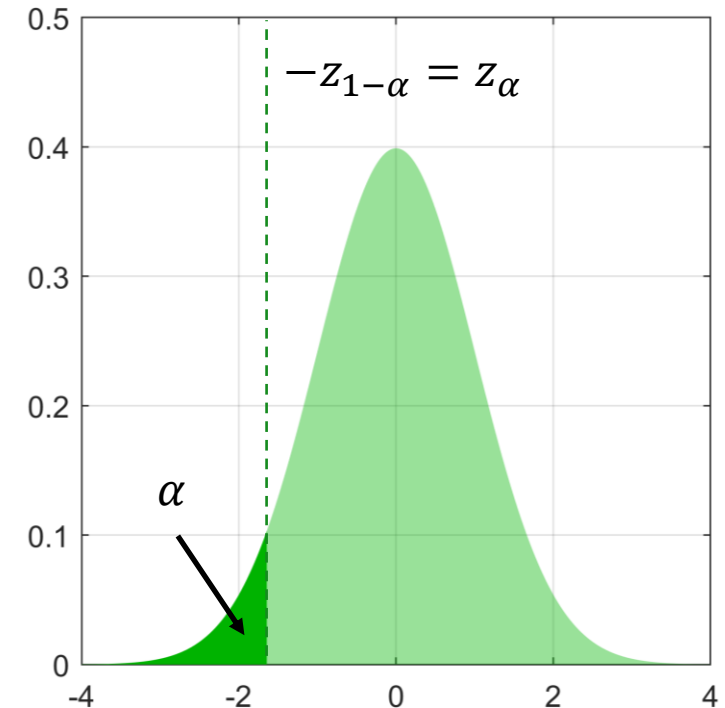
Rejection region

$$p = \Phi(z)$$

P-value

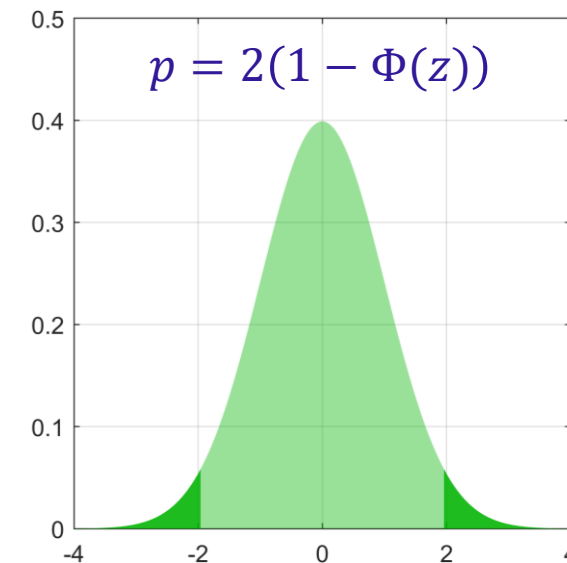
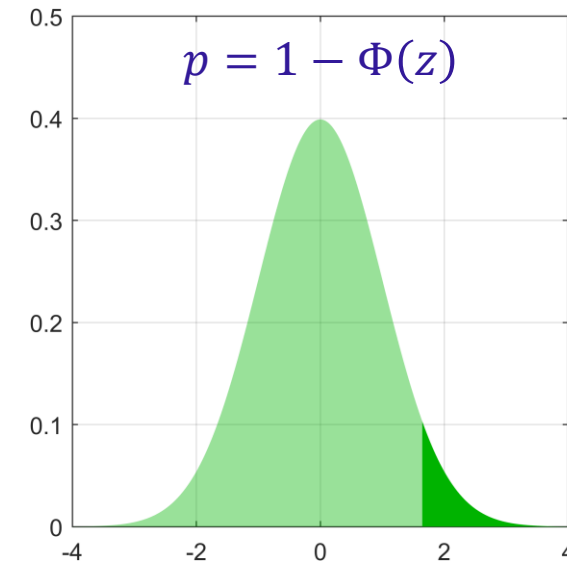


$$-z_{1-\alpha} = z_{crit}$$



Choosing 1 tailed or 2 tailed

- It seems that 1 tailed would make sense in the case of the WBC treatment
- We want the drug to be better and 1 tailed increases the probability of rejecting H_0 for a given α
 - This would make the rejection region on the right larger, or the make the p-value smaller
 - Choosing 1 tailed test can be interpreted as cheating or '*p-hacking*'
- 1 tailed is only proper if we don't care what happens with the other tails
 - e.g. if the drug reduces WBC count, we would care
- By convention, use 2 tailed unless you have an excellent justification and are prepared to defend it!
 - Some examples in the exercises





NHST outcomes

Types of Error and Power

NHST outcomes

H_0	True	False
Do not reject	True negative 	False negative  Type II error
Reject	False positive  Type I error	True positive 

‘negative’ here means ‘no effect’
i.e. the null hypothesis is true

- Analogous to the presumption of innocence: innocent until proven guilty
 - H_0 : innocent
 - H_1 : guilty
- Sometimes an innocent person is found to be guilty – Type I error
- Sometimes a guilty person is found to be innocent – Type II error
- We must make trade-off between the two



NHST outcomes and power

H ₀	True	False
Do not reject	True -ve	False -ve Type II
Reject	False +ve Type I	True +ve

H ₀	True	False
Do not reject	1 - α	β
Reject	α	1 - β

- α is the probability of a Type 1 error
 - False positive
 - $P(\text{Reject } H_0 | H_0) = \alpha$
 - e.g. α = 0.05*
 - The drug doesn't work, but you think it does
 - Innocent, but found guilty
- 1 - α is the probability of true negative
 - $P(\text{Accept } H_0 | H_0) = 1 - \alpha$
- β is the probability of a Type 2 error
 - False negative
 - $P(\text{Accept } H_0 | H_1) = \beta$
 - e.g. β = 0.2*
 - The drug works, but you think it doesn't
 - Guilty, but found innocent
- 1 - β is the probability of a true positive
 - $P(\text{Reject } H_0 | H_1) = 1 - \beta$
 - This is called Statistical **power**
 - Tells us how likely we are to detect an effect *if it exists*
 - Trade off with the risk of false negatives

Of the child whiche kepte the sheep

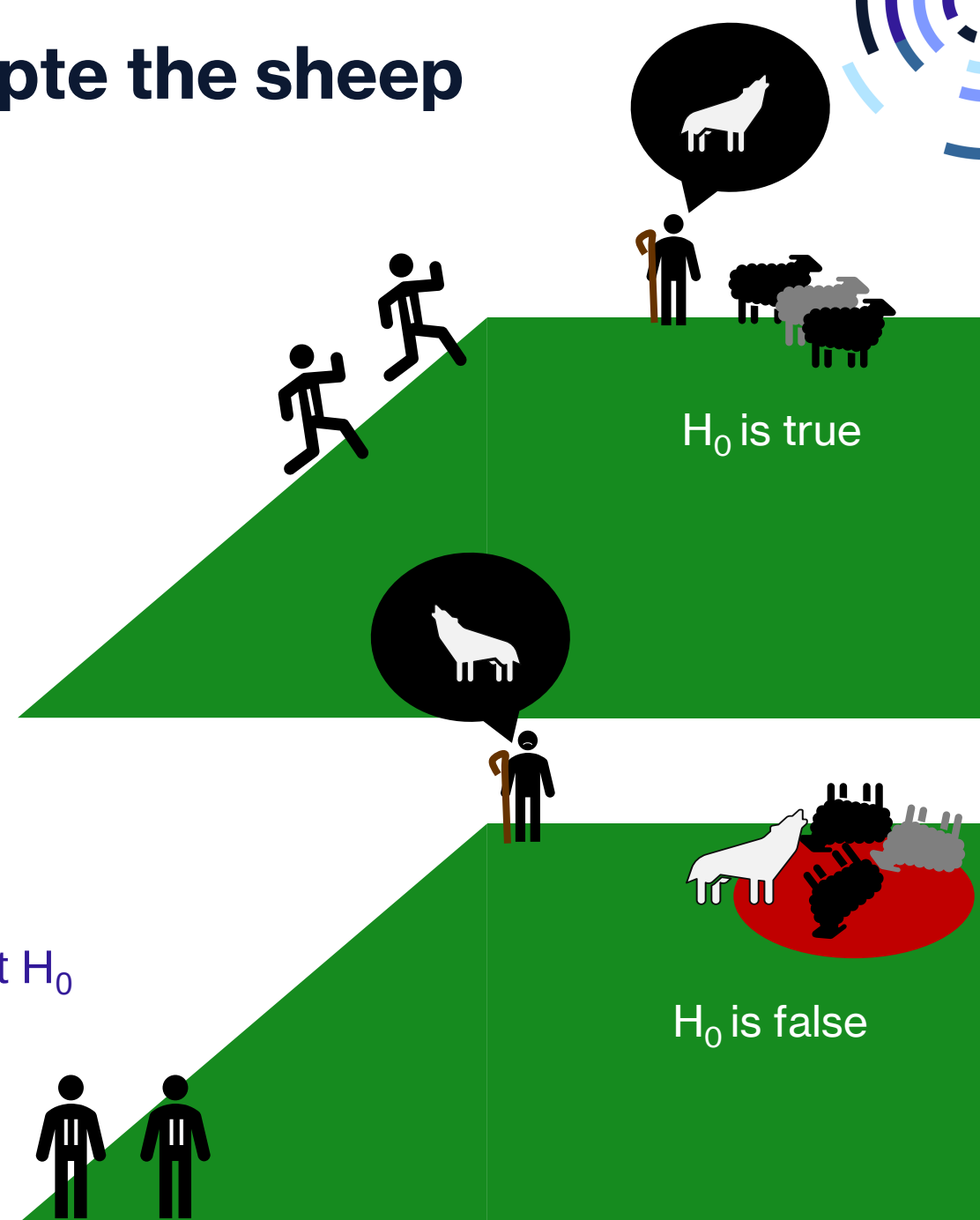
What Caxton called 'The boy who cried wolf' in 1484 (thanks Wikipedia)

1. The boy was bored and shouted that there was a wolf for fun
2. The villagers came to help, but there was no wolf
3. Later there was a wolf and the boy shouted but the villagers didn't show
4. The Sheep got got

H_0 : there is no wolf

Type I: villagers incorrectly reject H_0

Type II: villagers incorrectly do not reject H_0





Probability of Type I and Type II Errors

False Positives and False Negatives

If H_0 is true

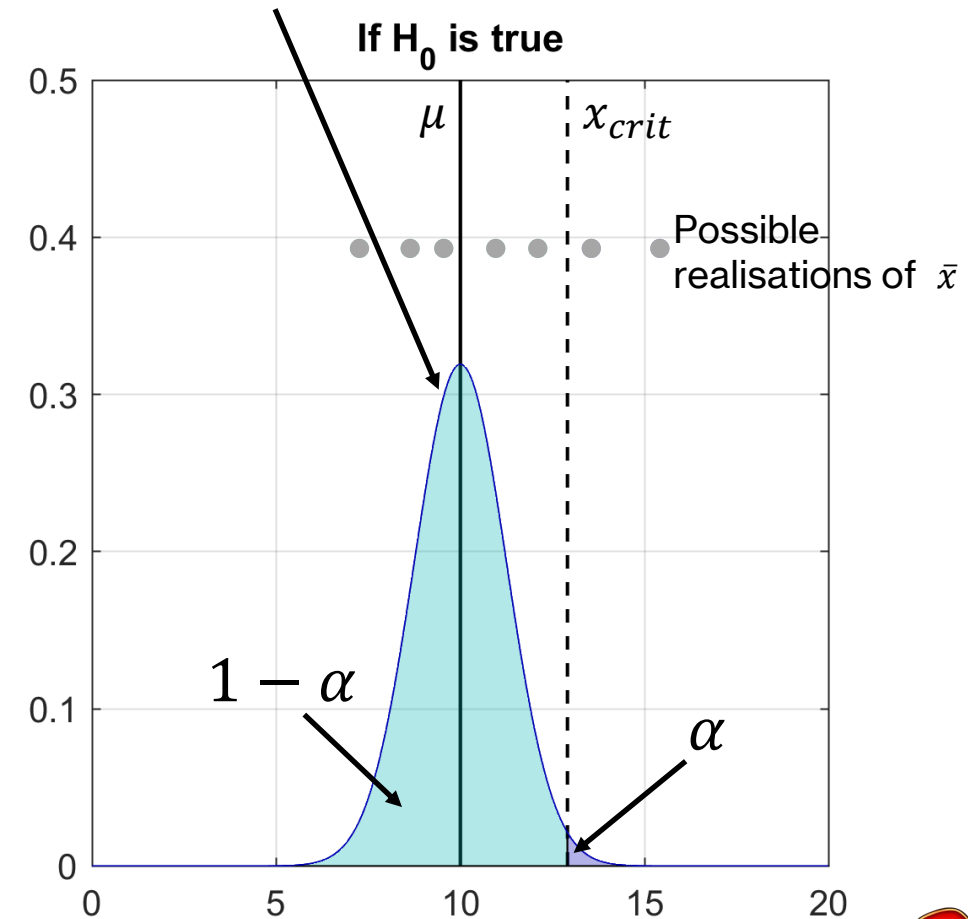
- Consider upper tailed test **if H_0 is true**
- We want to compare a sample mean to a value $\mu = 10$, and a measurement will yield $\bar{x}$ and we know $\sigma_{\bar{X}}$
- We calculate z and compare to $z_{1-\alpha}$
- In general, the limit of the rejection regions is known as z_{crit}
- Equivalently, we could calculate x_{crit}

$$x_{crit} = z_{crit}\sigma_{\bar{X}} + \mu_0$$

$$z = \frac{\bar{x} - \mu_0}{\sigma_{\bar{X}}}$$

- If $\bar{x} < x_{crit}$, we have a true negative
 - $P(\bar{x} < x_{crit} | H_0) = 1 - \alpha$
- If $\bar{x} > x_{crit}$, we have a false positive
 - $P(\bar{x} > x_{crit} | H_0) = \alpha$

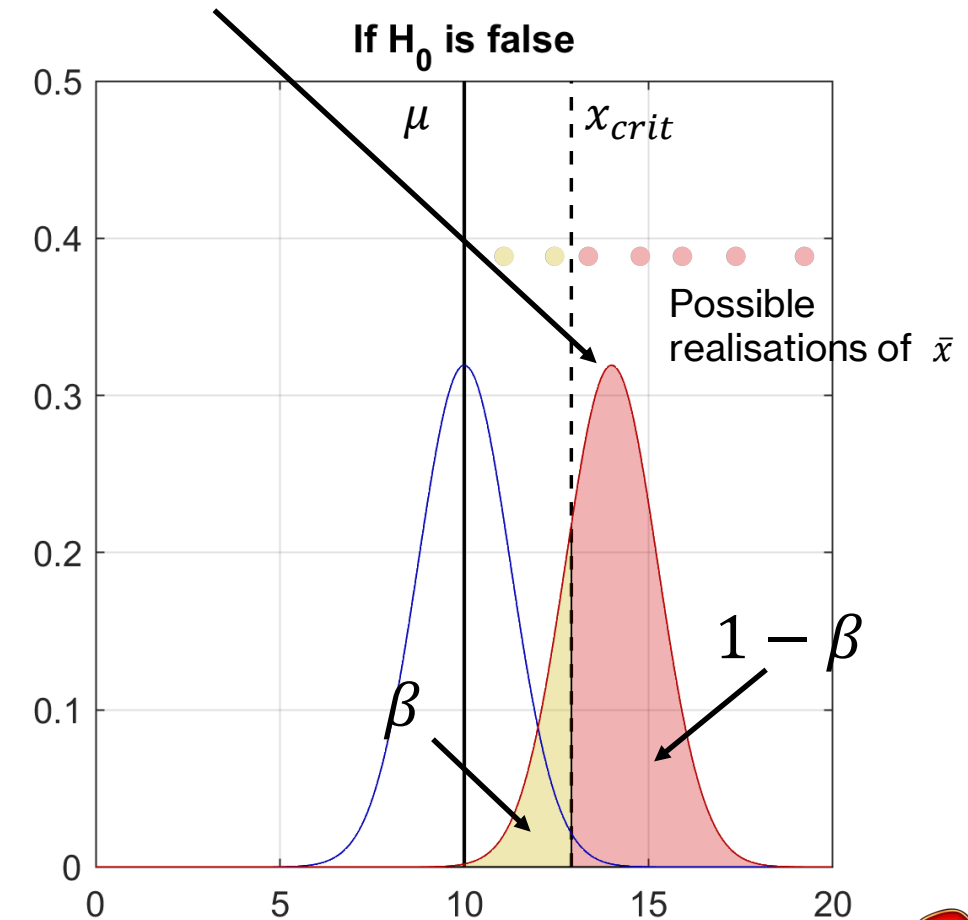
Distribution of $\bar{x}$ if H_0 is true



If H_0 is false

- Consider the same upper-tailed test **if H_0 is false**
- A measurement will yield $\bar{x}$
- If $\bar{x} < x_{crit}$, we have a false negative
 - $P(\bar{x} < x_{crit} | H_1) = \beta$
- If $\bar{x} > x_{crit}$, we have a true positive
 - $P(\bar{x} > x_{crit} | H_1) = 1 - \beta$
- Power is typically reported as a percentage
- Power = $(1 - \beta) \times 100\%$
- We want to maximise power

Distribution of $\bar{x}$ if H_0 is false



Increasing Power

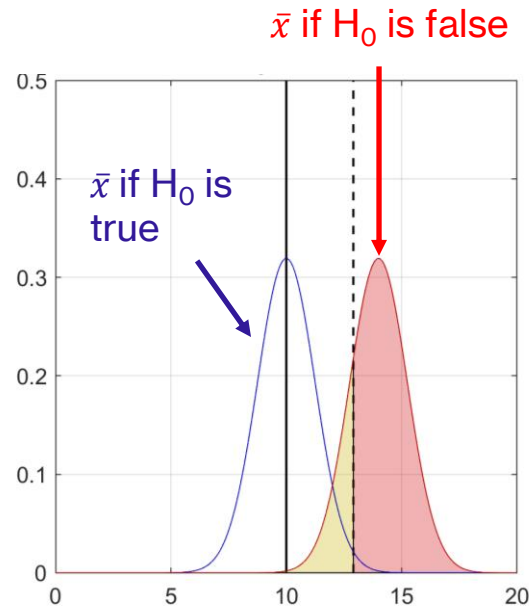
Default

$$\alpha = 0.01$$

$$\mu_1 - \mu_0 = 4$$

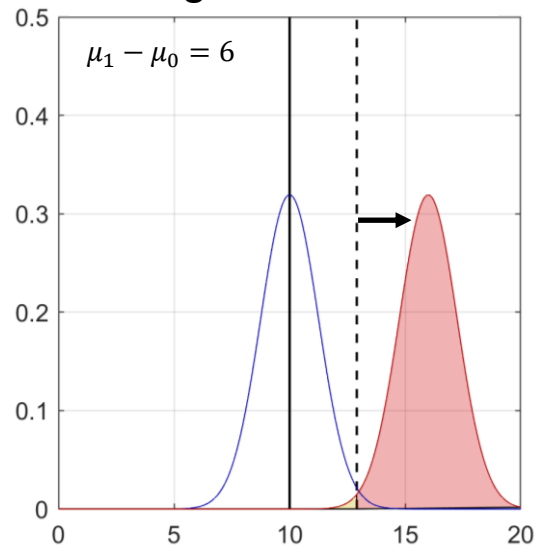
$$n = 16$$

$$\sigma = 5$$

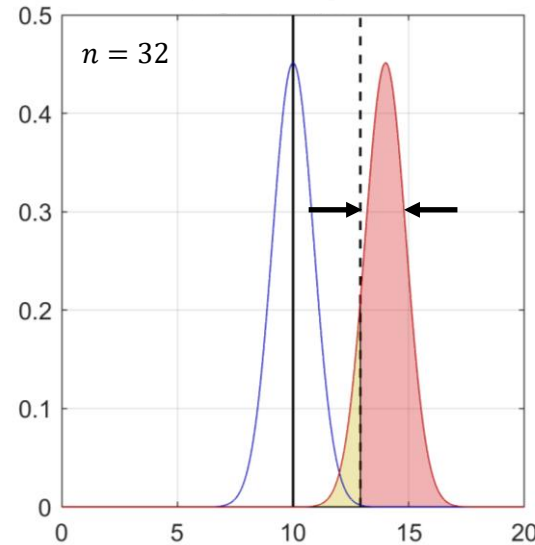


- We can increase power if
 - Effect size (difference between $\bar{x}$ and μ) increases, e.g. 6 vs 4
 - We have a larger sample size (32 vs 16)
 - Or a smaller standard deviation
 - We increase α (0.05 vs 0.01) 95% CI instead of 99% CI

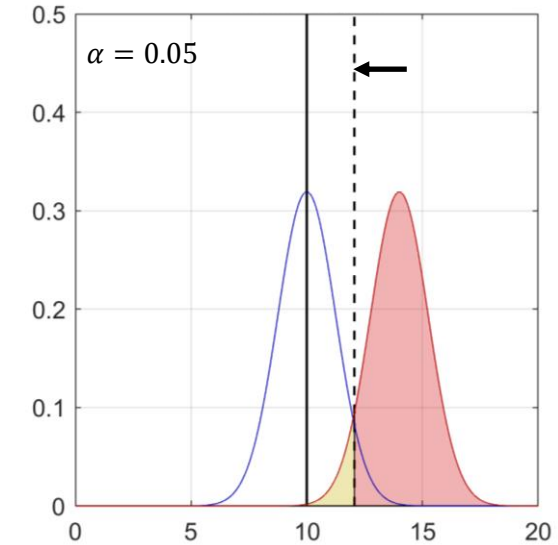
Larger effect size



Larger sample size



Larger alpha



Sample Size (upper-tailed z-test)

- When planning experiments, it is useful to be able to ask how many samples are needed
- For this, we use a sample size or power calculation
- For the case we have just looked at (upper tailed, z-test) there is an analytical solution (see next slide) for the power as a function of sample size, and vice versa

$$\beta = \Phi \left(z_{1-\alpha} + \frac{\mu_0 - \mu_1}{\sigma/\sqrt{n}} \right)$$

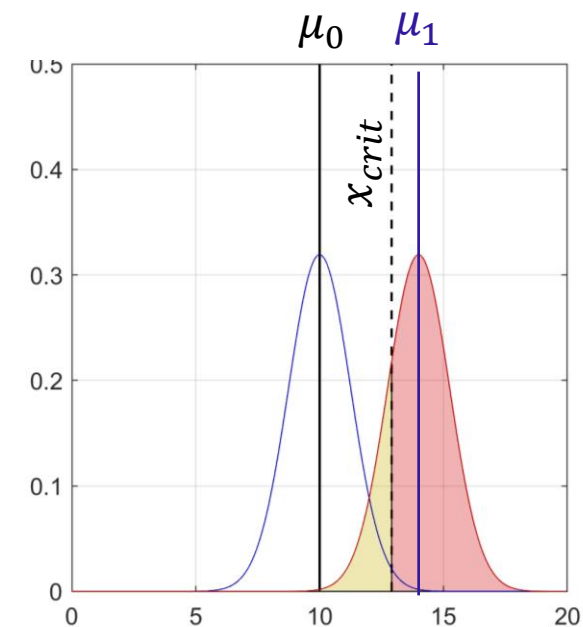
$$n = \left[\frac{\sigma(z_{1-\alpha} + z_{1-\beta})}{\mu_0 - \mu_1} \right]^2$$

- Note, both require knowledge of μ_1 and σ
- In practice, we tend to predict a value of μ_1 and based an estimate of σ on prior knowledge

FYI - Power Calculation

- To derive power calculation for 1-tailed z-test: H_0 is that $\mu_1 \leq \mu_0$
- Let $\mu_1 > \mu_0$
- Hence H_0 is false
- How likely is it that we get a Type II error?

β is the probability of a type II error or false negative



Standardise by μ_1 (subtract μ_1 and divide by $\sigma_{\bar{X}}$)

$$\beta(\mu_1) = P(H_0 \text{ is not rejected})$$

$$\beta(\mu_1) = P(\bar{X}_1 \leq X_{crit})$$

Recalling $X_{crit} = Z_{1-\alpha}\sigma_{\bar{X}} + \mu_0$

$$\beta(\mu_1) = P(\bar{X}_1 \leq Z_{1-\alpha}\sigma_{\bar{X}} + \mu_0)$$

$$\beta(\mu_1) = P\left(\frac{\bar{X}_1 - \mu_1}{\sigma_{\bar{X}}} \leq \frac{Z_{1-\alpha}\sigma_{\bar{X}} + \mu_0 - \mu_1}{\sigma_{\bar{X}}}\right)$$

rearrange

$$\beta(\mu_1) = P\left(\frac{\bar{X}_1 - \mu_1}{\sigma_{\bar{X}}} \leq Z_{1-\alpha} + \frac{\mu_0 - \mu_1}{\sigma_{\bar{X}}}\right)$$

$\frac{\bar{X}_1 - \mu_1}{\sigma_{\bar{X}}}$ is standardised $\bar{X}_1$, so use standard normal: $P(Z \leq z) = \Phi(z)$

$$\beta(\mu_1) = \Phi\left(Z_{1-\alpha} + \frac{\mu_0 - \mu_1}{\sigma_{\bar{X}}}\right)$$

FYI - Sample Size

- What about the other way around?
- We choose a power $(1 - \beta)$ and we want to know how many samples we need

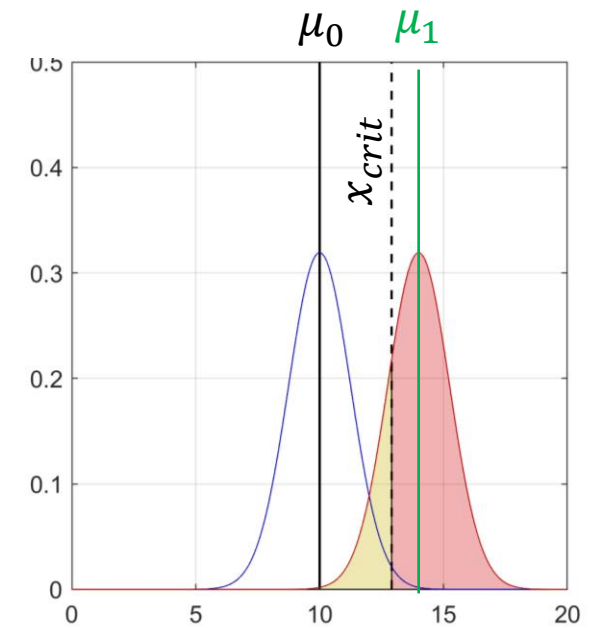
$$\beta(\mu_1) = \Phi\left(Z_{1-\alpha} + \frac{\mu_0 - \mu_1}{\sigma_{\bar{X}}}\right)$$

$$\Phi^{-1}(1 - \beta) = Z_{1-\beta} = Z_{1-\alpha} + \frac{\mu_0 - \mu_1}{\sigma_{\bar{X}}}$$

$$Z_{1-\beta} - Z_{1-\alpha} = \frac{\mu_0 - \mu_1}{\sigma_{\bar{X}}}$$

$$\sigma_{\bar{X}}(Z_{1-\beta} - Z_{1-\alpha}) = \mu_0 - \mu_1$$

β is the probability of a type II error or false negative



$$\frac{\sigma_X(Z_{1-\beta} - Z_{1-\alpha})}{\mu_0 - \mu_1} = \sqrt{n} \quad \sigma_{\bar{X}} = \frac{\sigma_X}{\sqrt{n}}$$

$$n = \left[\frac{\sigma_X(Z_{1-\beta} - Z_{1-\alpha})}{\mu_0 - \mu_1} \right]^2$$

FYI - Sample Size

- What about the other way around?
- We choose a power $(1 - \beta)$ and we want to know how many samples we need. Start with β

$$\beta(\mu_1) = \Phi\left(Z_{1-\alpha} + \frac{\mu_0 - \mu_1}{\sigma_{\bar{X}}}\right)$$

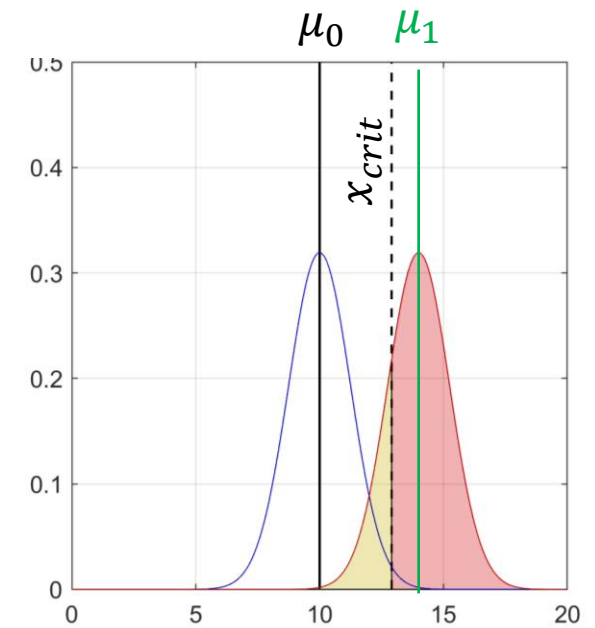
$$\Phi^{-1}(\beta) = Z_{\beta} = Z_{1-\alpha} + \frac{\mu_0 - \mu_1}{\sigma_{\bar{X}}}$$

$$Z_{1-\beta} = -Z_{\beta}$$

$$-Z_{1-\beta} - Z_{1-\alpha} = \frac{\mu_0 - \mu_1}{\sigma_{\bar{X}}}$$

$$-\sigma_{\bar{X}}(Z_{1-\beta} + Z_{1-\alpha}) = \mu_0 - \mu_1$$

β is the probability of a type II error or false negative



$$\frac{-\sigma_X(Z_{1-\beta} + Z_{1-\alpha})}{\mu_0 - \mu_1} = \sqrt{n}$$

$$\sigma_{\bar{X}} = \frac{\sigma_X}{\sqrt{n}}$$

$$n = \left[\frac{\sigma_X(Z_{1-\beta} + Z_{1-\alpha})}{\mu_0 - \mu_1} \right]^2$$

Minus sign cancelled by the square

Sample Size (2-tailed z-test)

- For the case of a two tailed, z-test there is an approximate solution

Upper-tailed

$$\beta = \Phi \left(z_{1-\frac{\alpha}{2}} - \frac{|\mu_0 - \mu_1|}{\sigma/\sqrt{n}} \right)$$

$$\beta = \Phi \left(z_{1-\alpha} + \frac{\mu_0 - \mu_1}{\sigma/\sqrt{n}} \right)$$

$$n = \left[\frac{\sigma \left(z_{1-\frac{\alpha}{2}} + z_{1-\beta} \right)}{\mu_0 - \mu_1} \right]^2$$

$$n = \left[\frac{\sigma (z_{1-\alpha} + z_{1-\beta})}{\mu_0 - \mu_1} \right]^2$$

Power and Sample Size Example

- Find the power for the upper-tailed test considered in this section for $\alpha = 0.01$ and the sample size required for a power of 90%

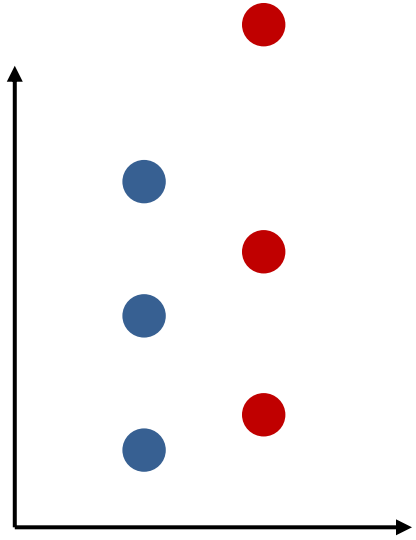
Power for $n = 16$

$$\beta(\mu_1) = \Phi\left(z_{1-\alpha} + \frac{\mu_0 - \mu_1}{\sigma/\sqrt{n}}\right)$$

$$\begin{aligned}\mu_1 - \mu_0 &= 4 \\ n &= 16 \\ \sigma &= 5\end{aligned}$$

Sample Size (n) for power of 90%

$$n = \left[\frac{\sigma(z_{1-\alpha} + z_{1-\beta})}{\mu_0 - \mu_1} \right]^2$$



The independent two-sample Z-test

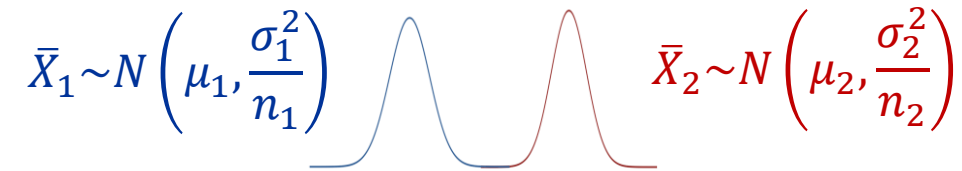
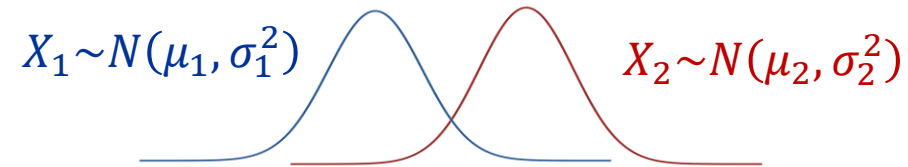
Analysing two unpaired samples



Two Independent Samples

- To this point, we have assumed that we know the population mean of the untreated population
- If this is not the case, we need to measure a sample of untreated patients to get $\bar{x}_1$ as an estimator of μ_1
 - Now we are dealing with a statistic instead of a known mean, I'm going to rename the control RV be X_1 and let the treated RV be X_2
- We will need to take into account the precision of our estimate of $\bar{x}_1$
- To do so, we can define a new RV $\bar{Y}$: the difference between the sample means

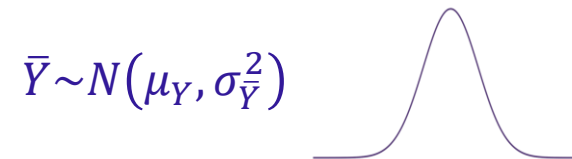
$$\bar{Y} = \bar{X}_2 - \bar{X}_1$$



- We know that

$$\mu_{\bar{Y}} = \mu_2 - \mu_1$$

$$\sigma_{\bar{Y}}^2 = \frac{\sigma_2^2}{n_2} + \frac{\sigma_1^2}{n_1}$$



For 2 unpaired samples

$$\bar{Y} = \bar{X}_2 - \bar{X}_1$$

$$\sigma_{\bar{Y}} = \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$$

- We now just treat $\bar{Y}$ like we did previously for $\bar{X}_1$

WBC Count Confidence Interval (2 samples)

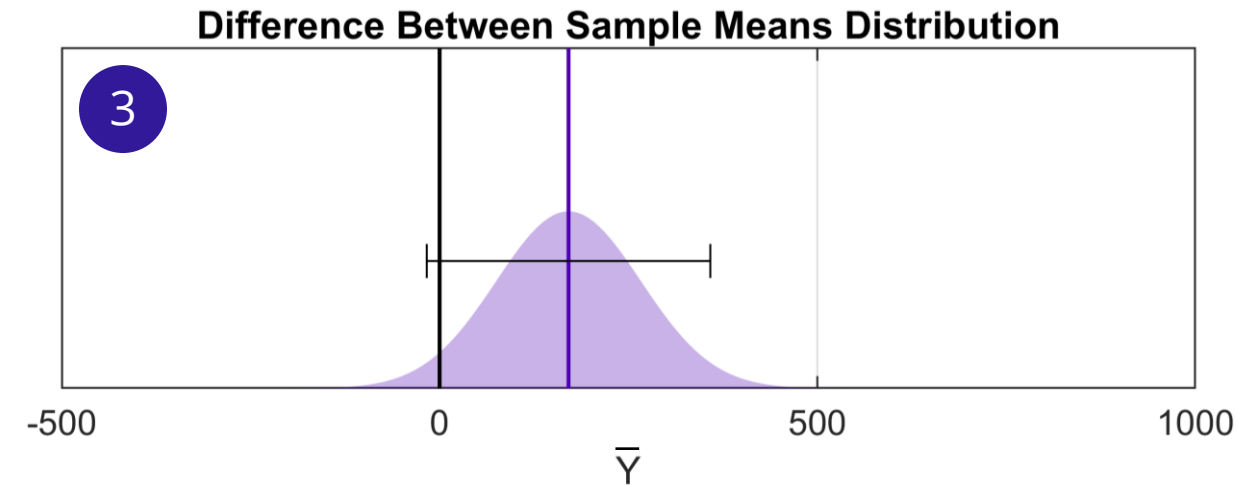
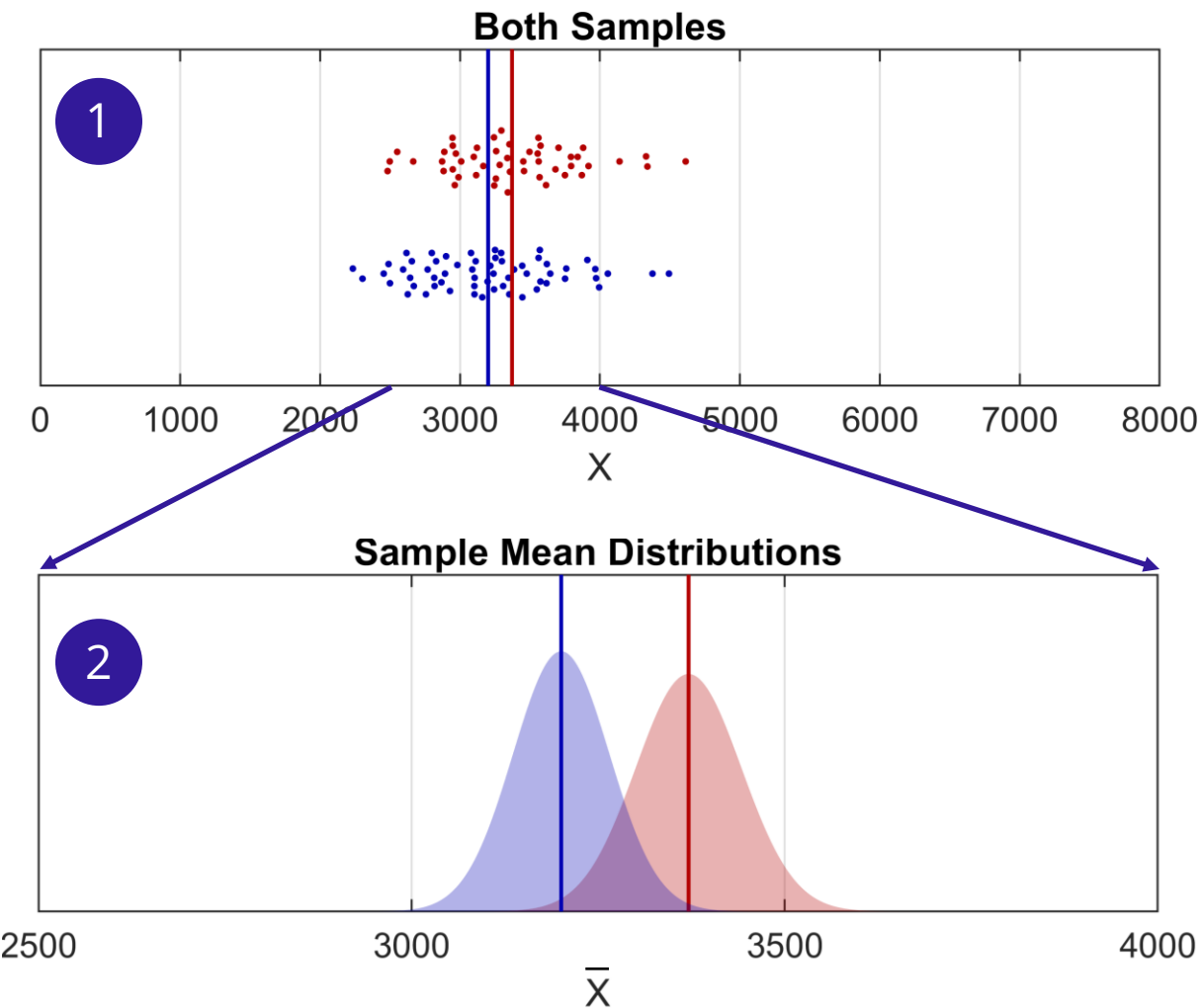


- X = number of white blood cells
- Clinical trial yields $n = 60$ values of X_1 for untreated individuals with $\bar{x}_1 = 3,201$
- Clinical trial yields $n = 50$ values of X_2 for treated individuals with $\bar{x}_2 = 3,371$
- Assume that the treatment doesn't affect the SD, hence $\sigma_2 = \sigma_1 = 500$
- From these values, we can calculate $\bar{y}$

Using $\alpha = 0.05$ (95% confidence interval), is it likely that we observed this value of $\bar{y}$ if $\mu_y = 0$?

- 0 is in the 95% confidence interval on $\bar{y}$
- If $\mu_2 = \mu_1$, we would expect this to occur 95 in 100 times
- Let's visualise

WBC Count Visualisation

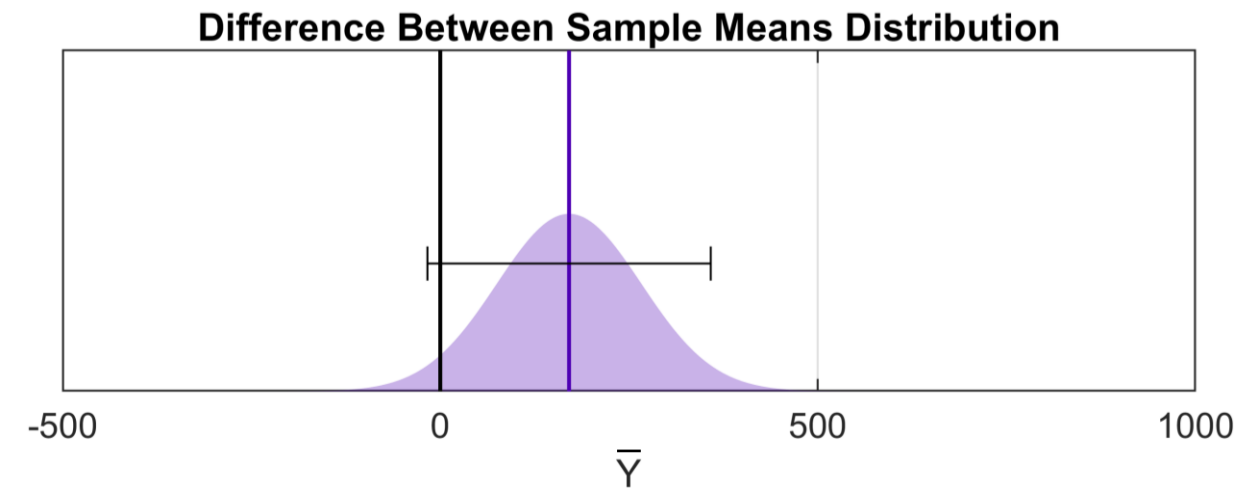
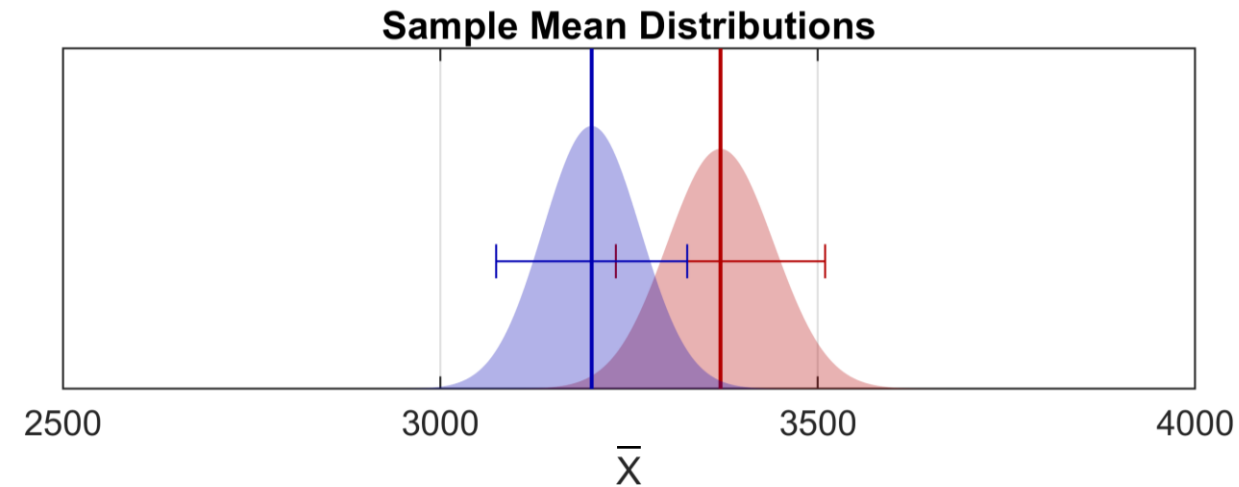


- If $\mu_2 = \mu_1$, we would expect the CI on $\bar{y}$ to contain zero 95 in 100 times
- We conclude that
- “ μ_2 is not statistically different to μ_1 ”

Did we need confidence interval on $\bar{Y}$



- Could we have taken a shortcut and just checked if the confidence intervals on $\bar{X}_2$ and $\bar{X}_1$ overlapped?
- If CI_1 and CI_2 **do not** overlap then CI_Y **will not** include zero
- If CI_1 and CI_2 **do** overlap then CI_Y **may or may not** include zero
- So technically, only if there is an overlap do we need to check



FYI – no CI overlap proof

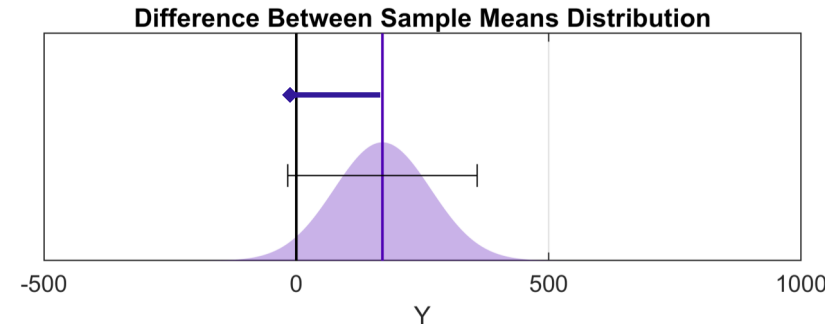
$$\bar{Y} = \bar{X}_2 - \bar{X}_1, \sigma_{\bar{Y}} = \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}, \text{ consider for } \bar{Y} > 0 \text{ or } \bar{X}_2 > \bar{X}_1,$$

Statement 1

CI_Y will not include zero if

- $\bar{Y} - z_{crit}\sigma_{\bar{Y}} > 0$
- $\bar{Y} > z_{crit}\sigma_{\bar{Y}}$
- $\bar{X}_2 - \bar{X}_1 > z_{crit}\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$

1



Square term 1

$$\left(\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}} \right)^2 = \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}$$

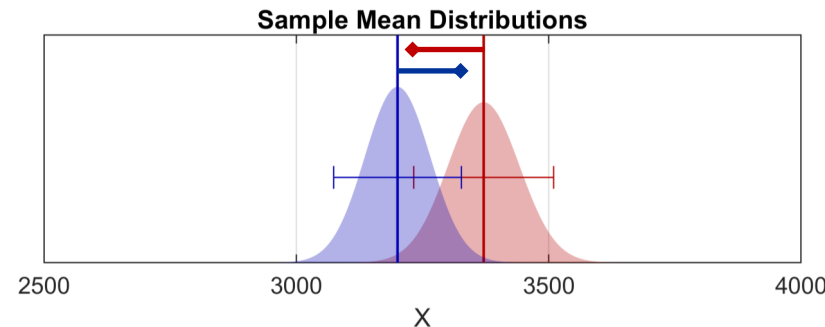
1

Statement 2

For no overlap between MEs on X

- $\bar{X}_2 - z_{crit}\sqrt{\frac{\sigma_2^2}{n_2}} > \bar{X}_1 + z_{crit}\sqrt{\frac{\sigma_1^2}{n_1}}$
- $\bar{X}_2 - \bar{X}_1 > z_{crit} \left(\sqrt{\frac{\sigma_1^2}{n_1}} + \sqrt{\frac{\sigma_2^2}{n_2}} \right)$

2



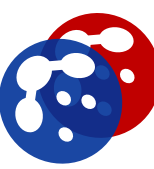
Square term 2

$$\left(\sqrt{\frac{\sigma_1^2}{n_1}} + \sqrt{\frac{\sigma_2^2}{n_2}} \right)^2 = \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2} + 2\sqrt{\frac{\sigma_2^2}{n_2}}\sqrt{\frac{\sigma_1^2}{n_1}}$$

2

Hence Term 2 is always greater than Term 1

If Statement 2 is true, Statement 1 will also be true, but not vice versa



WBC Null-Hypothesis Approach

- ‘Does the drug affect WBC count?’
 - H_0 : there is no difference between the population means of the treated and untreated populations

$$\mu_1 = \mu_2$$

or more generally for a reference value μ_0

$$\mu_2 - \mu_1 = \mu_0$$

- H_1 : there is a difference between the population means of the treated and untreated populations

$$\mu_2 - \mu_1 \neq \mu_0$$

or for $\mu_0 = 0$

$$\mu_1 \neq \mu_2$$

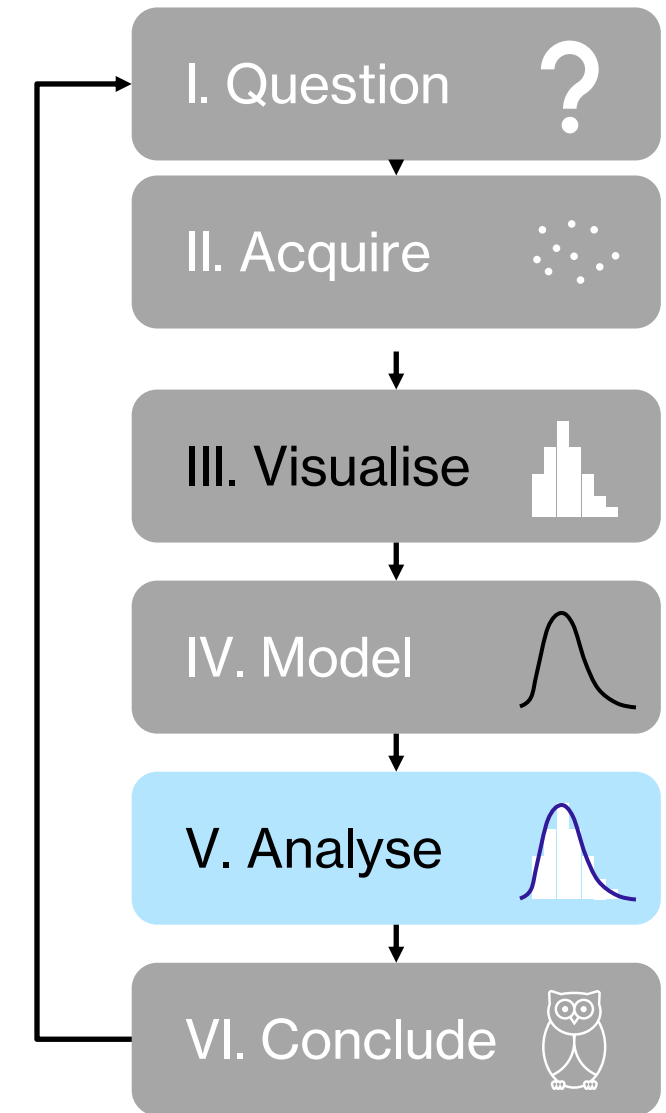
- Model: $\bar{Y} \sim N(\mu_Y, \sigma_{\bar{Y}}^2)$

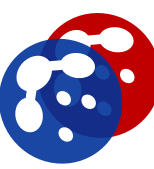
$$Z = \frac{\bar{Y} - \mu_0}{\sigma_{\bar{Y}}}$$

μ_0 is a known reference value

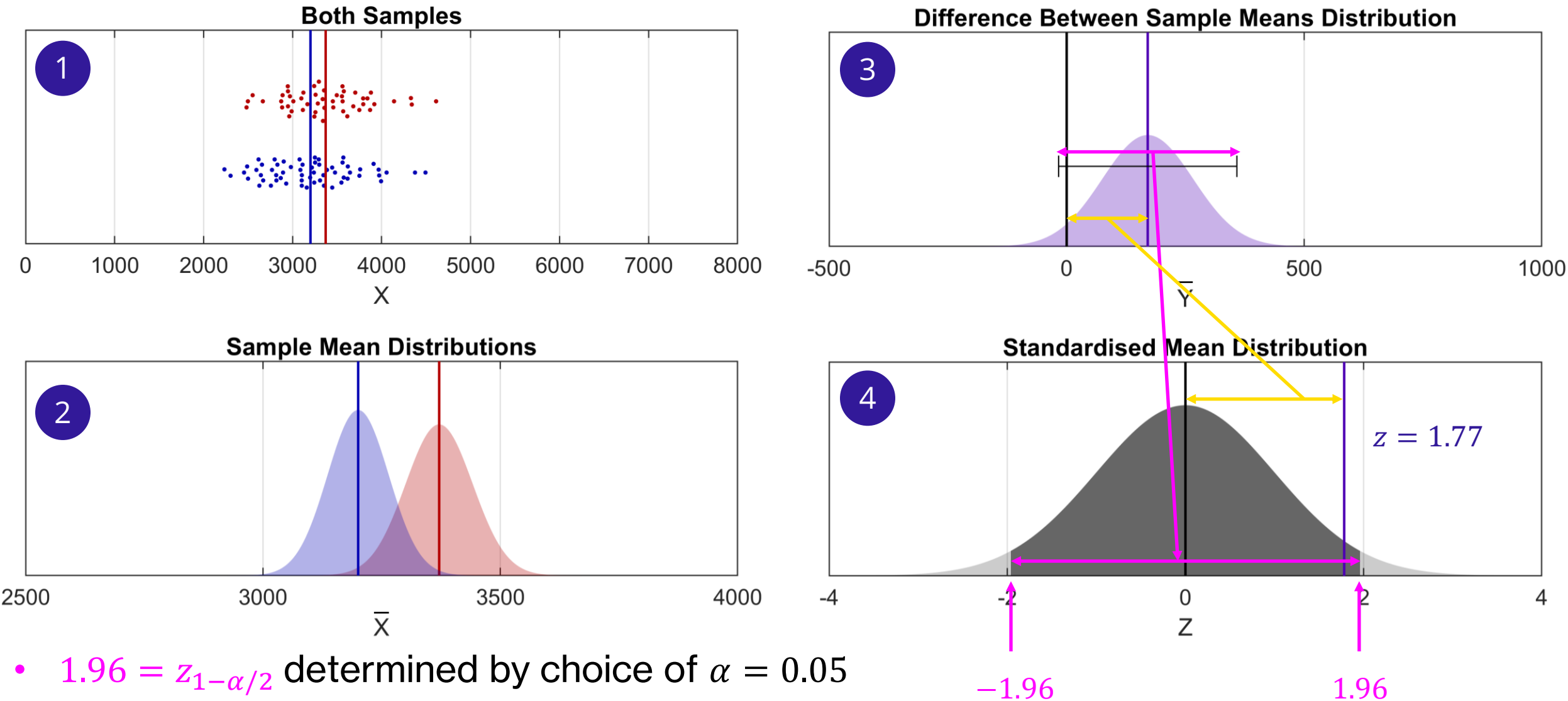
- Or specifically for this case

$$z = \frac{\bar{y} - 0}{\sigma_{\bar{Y}}} = \frac{170}{96} = 1.77$$

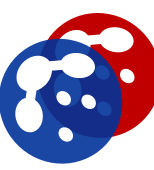




WBC Count Z-test Visualisation



- $1.96 = z_{1-\alpha/2}$ determined by choice of $\alpha = 0.05$
- z -statistic is determined by the data



WBC Count two sample unpaired Z-test

Using rejection region

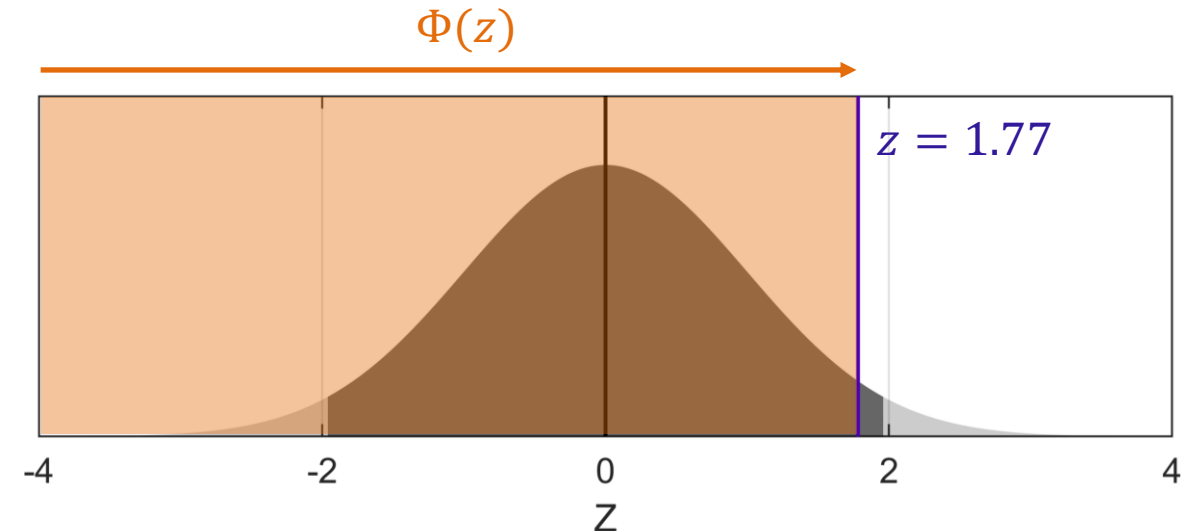
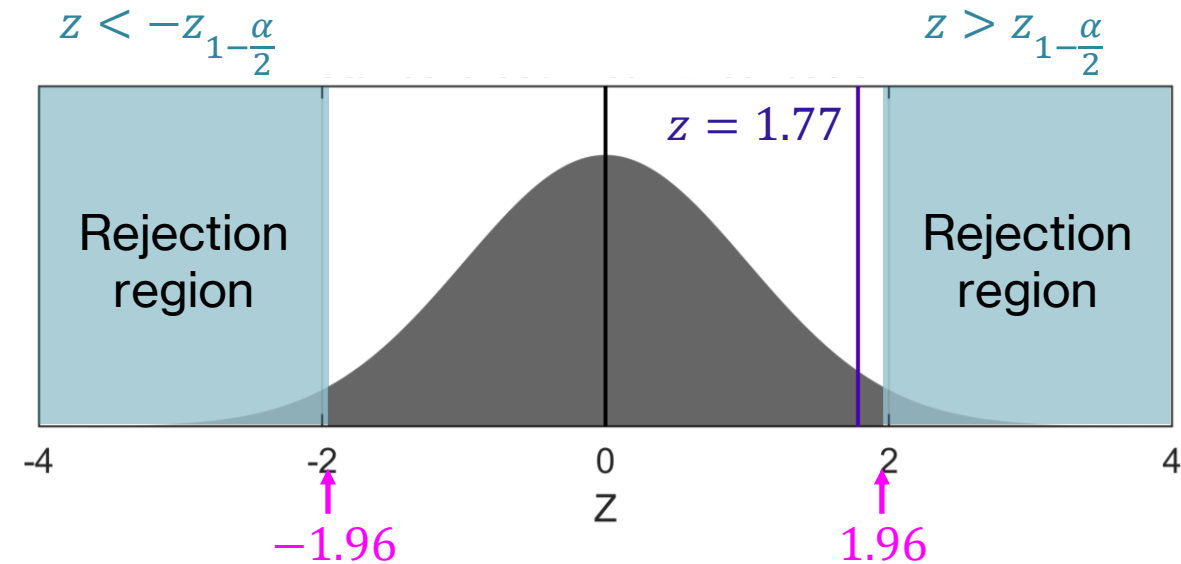
$$|z| > z_{1-\frac{\alpha}{2}}$$

- Our **z-statistic** is not in the rejection region, hence we “do not reject the null hypothesis”

Using p-value

$$p = 2(1 - \Phi(|1.77|))$$

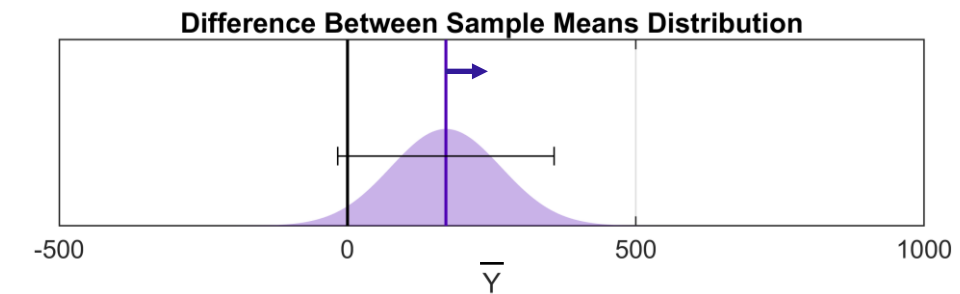
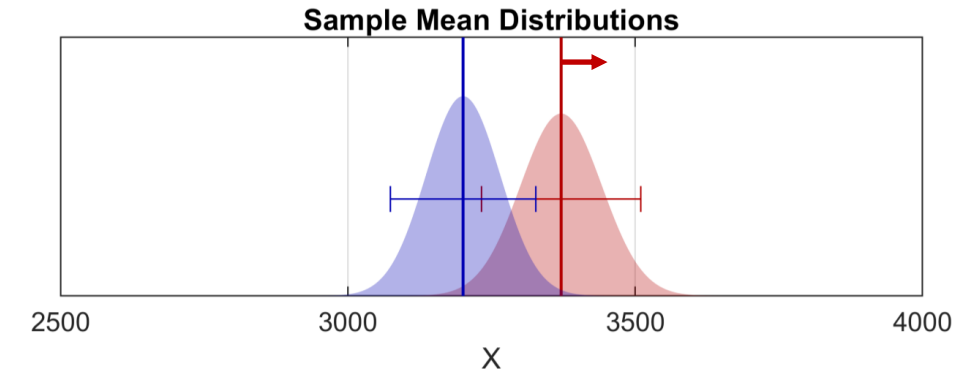
- $\Phi(z) = 0.9616$, $p = 0.077$
- As $p > \alpha$, we “do not reject H_0 ”



What can we learn?



- We cannot reject H_0 ($\mu_1 = \mu_2$) with (or 'at') $\alpha = 0.05$
- This **does not** mean that
 - “ H_0 is true”
 - “The treatment had no effect”
- It **does** mean that
 - “We could not reject H_0 based on the data”
 - “We did not detect a statistical difference due to the treatment”
- Can we learn anything beyond this?
 - ME_Y gives us an indication of the ‘detection limits’ of the measurement
 - For our $ME_Y = 188$ we could only have detected a difference of 188 or greater
- Why does this matter?
 - Firstly, if we had larger sample sizes, then $\sigma_{\bar{Y}}$ and hence ME_Y would have been smaller
 - The precision of our measurement would have improved and we may have detected a statistical difference
- Additionally, we might consider what contributes to $\sigma_{\bar{Y}}$



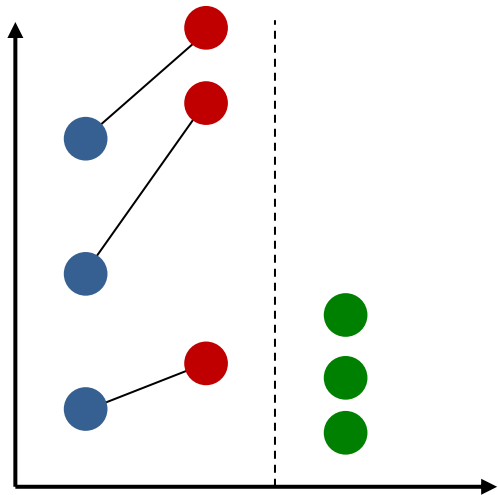
If H_0 is not rejected, we can use the ME/CI to investigate how big a change we could have detected



Considering multiple sources of uncertainty

- Let us consider the variable σ_1^2
- In real terms, this would be the variance in our **measured** WBC count for the untreated population
- What values contribute to σ_1^2 ?
 1. Variance in the population, σ_{pop}^2
 2. Variance in the measurement, σ_{meas}^2
- Knowing what we do about adding variances, we could model
$$\sigma_1^2 = \sigma_{pop,1}^2 + \sigma_{meas}^2$$
- Hence improving the precision of the measurement (reducing σ_{meas}^2) could reduce the variance on $\bar{x}$, reducing $\sigma_{\bar{Y}}$
- For the treated population, we may also consider that the treatment might affect different individuals differently (e.g. because patients had different disease states or genetics)

$$\sigma_2^2 = \sigma_{pop,2}^2 + \sigma_{meas}^2 + \sigma_{treat}^2$$



The dependent two-sample Z-test

Analysing two paired samples



Using paired samples

- Using the model proposed on the previous page, we could write

$$\sigma_Y^2 = \frac{\sigma_{pop,2}^2 + \sigma_{meas}^2 + \sigma_{treat}^2}{n_2} + \frac{\sigma_{pop,1}^2 + \sigma_{meas}^2}{n_1}$$

- What if we could remove one of these sources of variability?
 - We have 2 measurements, nothing we can do about that
 - We have the **treatment**, nothing we can do about that
 - We have $\sigma_{pop,1}^2$ the variability in the underlying population, nothing we can do about that
 - If we used the same individual for both experiments, we could eliminate $\sigma_{pop,2}^2$
 - Under the assumption that the same individual has the same WBC count

- In that case for each individual i , we could write

$$x_{2,i} = x_{1,i} + d_i$$

- Where d_i represents the change in WBC count due to the treatment for patient i

Two Dependent Samples



- We can rearrange for D

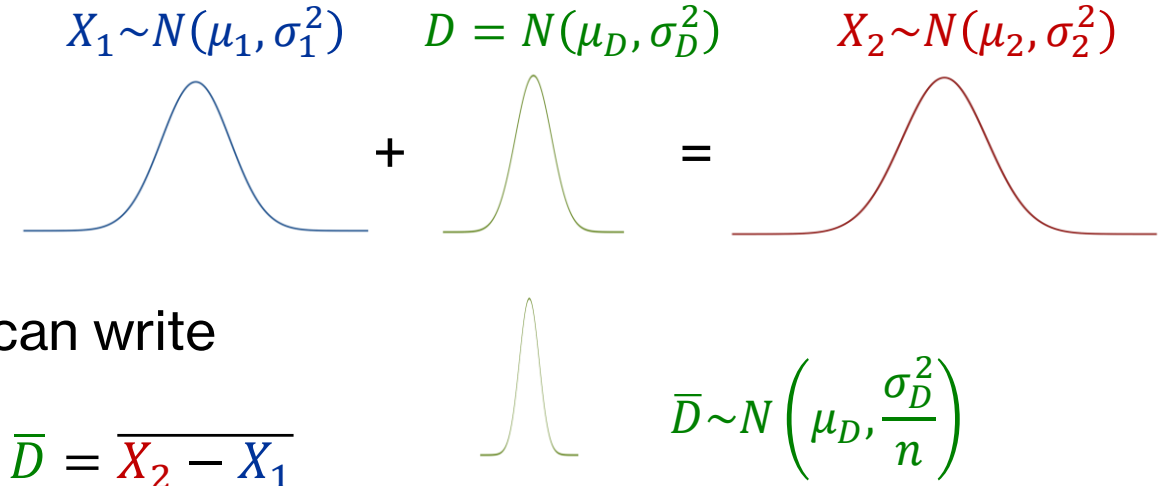
$$d_i = x_{2,i} - x_{1,i}$$
- Assuming that $x_{2,i}$ and $x_{1,i}$ are normally distributed, we get

$$D \sim N(\mu_D, \sigma_D^2)$$
- For now, we continue to assume that we know the population standard deviation, which is defined as

$$\sigma_D^2 = \frac{1}{N} \sum_{i=1}^N (d_i - \mu_D)^2$$

- For n paired samples, we can just calculate the mean and variance of the d_i values

$$\bar{d} = \frac{1}{n} \sum_{i=1}^n d_i$$



- We can write
- $\bar{D}$ is the **average** of the **differences**
- $\bar{Y}$ is the **difference** of the **averages**
- We now just do a 1-way test on $\bar{D}$
- Or could use CI on $\bar{D}$

For 2 paired samples

$$\bar{D} = \bar{X}_2 - \bar{X}_1$$

$$\sigma_{\bar{D}} = \frac{\sigma_D}{\sqrt{n}}$$

WBC Null-Hypothesis Approach



- ‘Does the drug affect WBC count?’
 - H_0 : the drug has no effect

$$\mu_d = 0$$
 or more generally for a reference value μ_0

$$\mu_d = \mu_0$$

- H_1 : there is a difference between the population means of the treated and untreated populations

$$\mu_d \neq \mu_0$$

or for $\mu_0 = 0$

$$\mu_d \neq 0$$

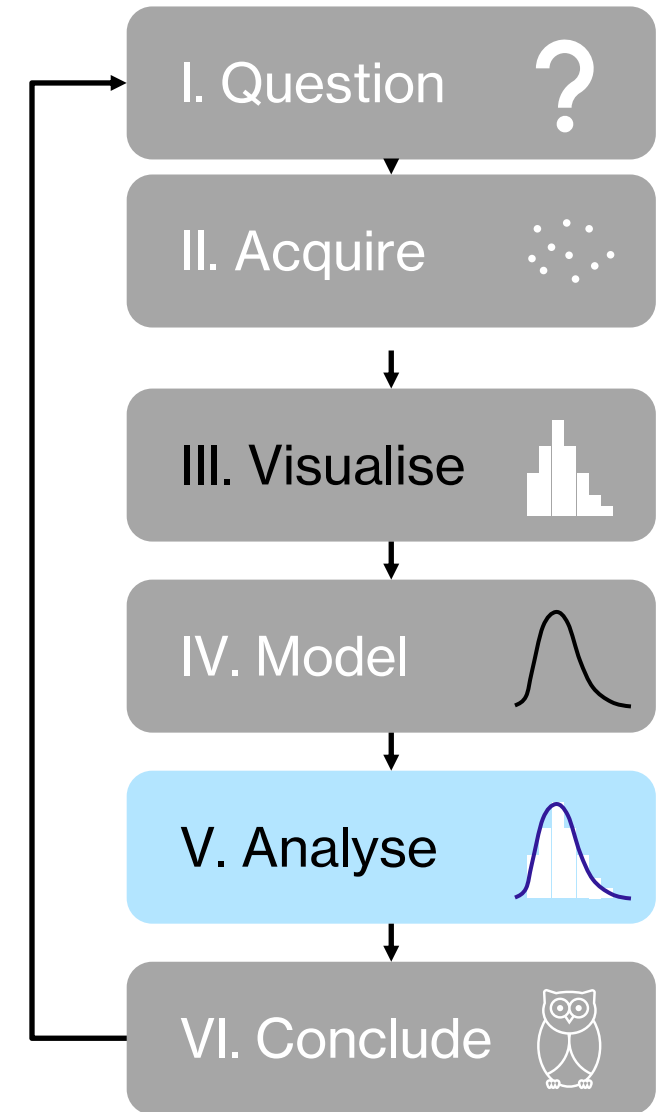
- Model: $\bar{D} \sim N(\mu_D, \sigma_D^2)$

$$Z = \frac{\bar{D} - \mu_0}{\sigma_{\bar{D}}}$$

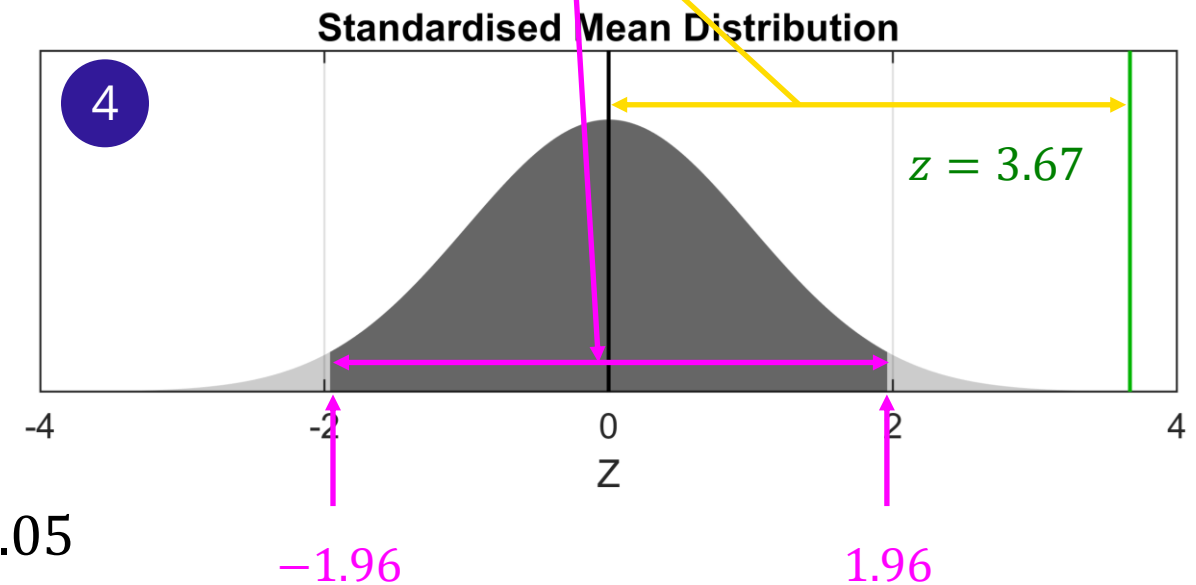
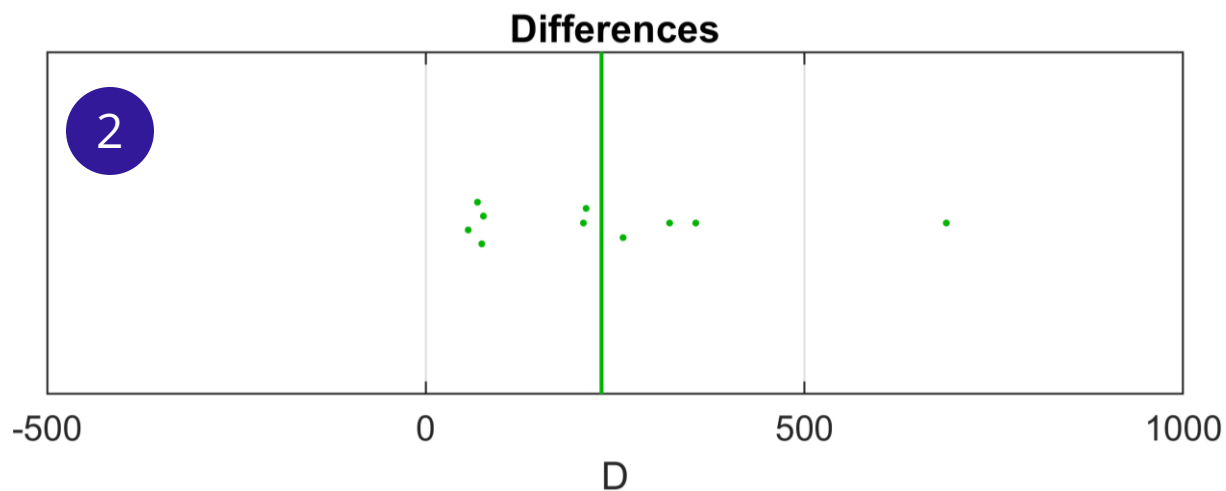
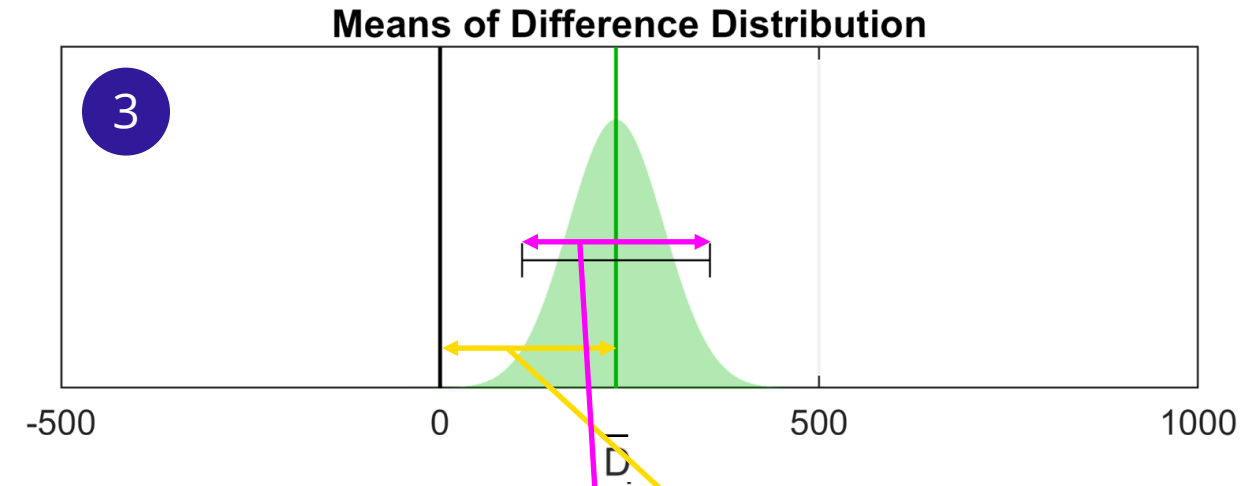
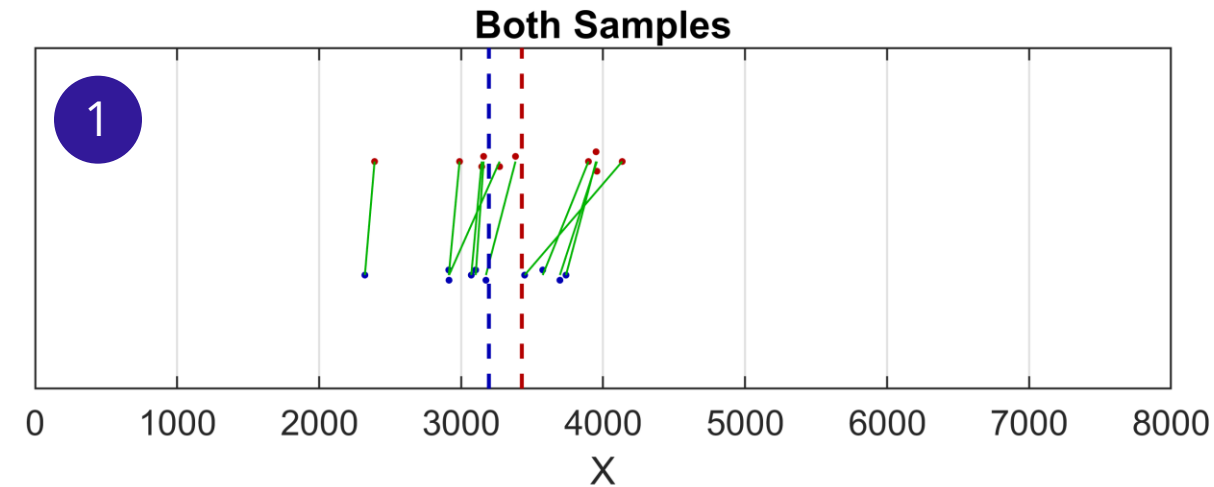
μ_0 is a known reference value

- Or specifically for this case

$$z = \frac{\bar{d} - 0}{\sigma_{\bar{D}}} = \frac{232}{63} = 3.67$$



WBC Count Z-test Visualisation



- $1.96 = z_{1-\alpha/2}$ determined by choice of $\alpha = 0.05$
- **z-statistic** is determined by the data

WBC Count two sample unpaired Z-test



Using rejection region

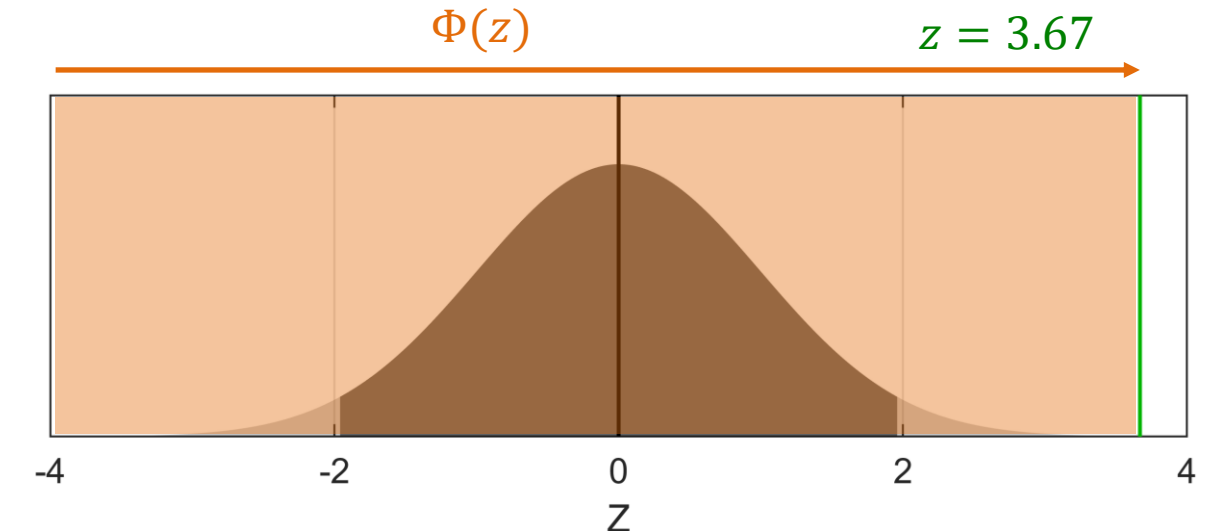
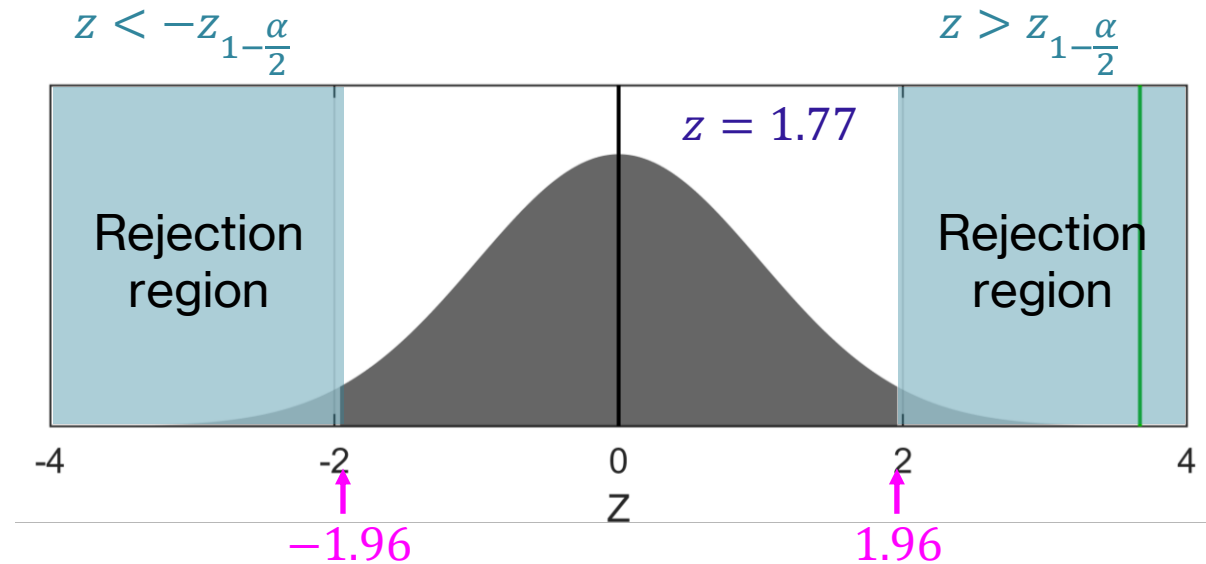
$$|z| > z_{1-\frac{\alpha}{2}}$$

- Our **z-statistic** is in the rejection region, hence we
“*reject the null hypothesis*”

Using p-value

$$p = 2(1 - \Phi(|3.67|))$$

- $\Phi(z) = 0.9999$, $p = 0.0002$
- As $p < \alpha$, we “*reject H_0* ”





Drawing proper conclusions from NHST

and the folly of interpreting exact p -values

Concluding

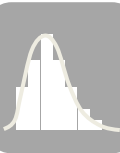
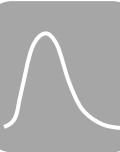
- For the paired analysis It would be common to conclude

*“The drug increased WBC count by 232, which was highly statistically
 3 significant ($p = 0.0002$)”* 4

- This would be wrong on many levels! Instead, we might say

*“The drug increased WBC count by an 1 average of 232 [108,356] 2
 4 ($p < 0.001$), although this only amounts to a
 7% increase over the untreated baseline values” 3*

- In the interest of clarity, we should explicitly state that the value is an average
- We must always report an average with its uncertainty (Lecture 5)
- Practical significance should be stated (7%=232/3200 from simulated data) and we don't use the term statistically significant (Slide 18)
- It is folly to read too much into exact p -values.....



Interpreting p -values

- The ' p -values' we calculated were an estimate of the probability of getting a test-statistic that is at least as extreme as we observed
- For example, for the paired sample, we got $p = 0.0002$
- Some would interpret this as
“we would only observe this outcome 1 in 5000 times, if there was no effect of the drug”
- They might therefore conclude that
“the effect of the drug was highly statistically significant”
- Let's run another Monte Carlo simulation to investigate whether it is reasonable to conclude a 'level' of significance from the p -value...
- We return to the wildlife conservationist



Interpreting p -values



- This time, let us consider that the conservationist has been feeding the badgers, and wants to evaluate if their weight has changed from $\mu_0 = 10$, using a one-sample, two-tailed test
- We would define
 H_0 : the average weight of the badgers has not changed
 H_1 : the average weight of the badgers has changed
- We run 100 simulated samples with `n_obs=10`
- For each one, we calculate
 - the 95% CI
 - the p-value and compare to $\alpha = 0.05$
- Code can be found in the `Lecture6_code.m`

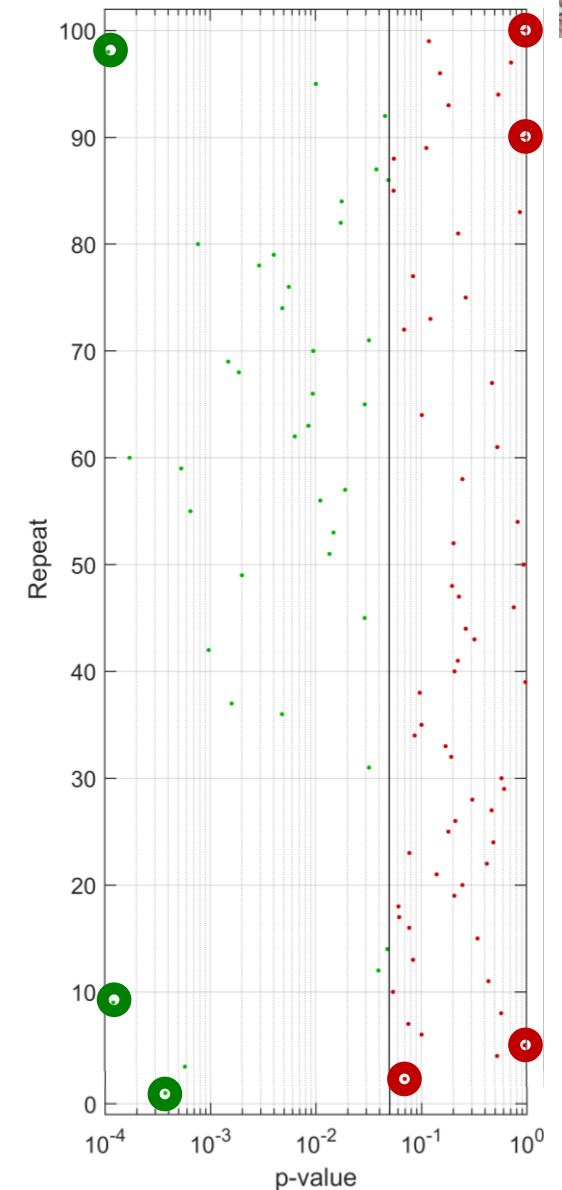


Interpreting p -values

- Our first sample yielded $p = 0.0004$ (1 in 2500)
- Our second sample yielded $p = 0.07$ (1 in 14)!
- We run 98 more samples and find p -values in the range 10^{-4} to 1!!!!

It is therefore, clearly folly to draw a strong conclusion from a single p -value

- R.A. Fisher is quoted as saying
“... if P is between 0.1 and 0.9 there is certainly no reason to suspect the hypothesis tested. If it's below 0.02 it is strongly indicated that the hypothesis fails to account for the whole of the facts. We shall not often be astray if we draw a conventional line at 0.05”
- Suffice it to say, lots of people disagree (do you?)
- In the current simulation 62 of 100 were above 0.05 and the remaining 38 were below
- The population value was actually $11kg$ (H_0 should be rejected), so we were astray more often than not!

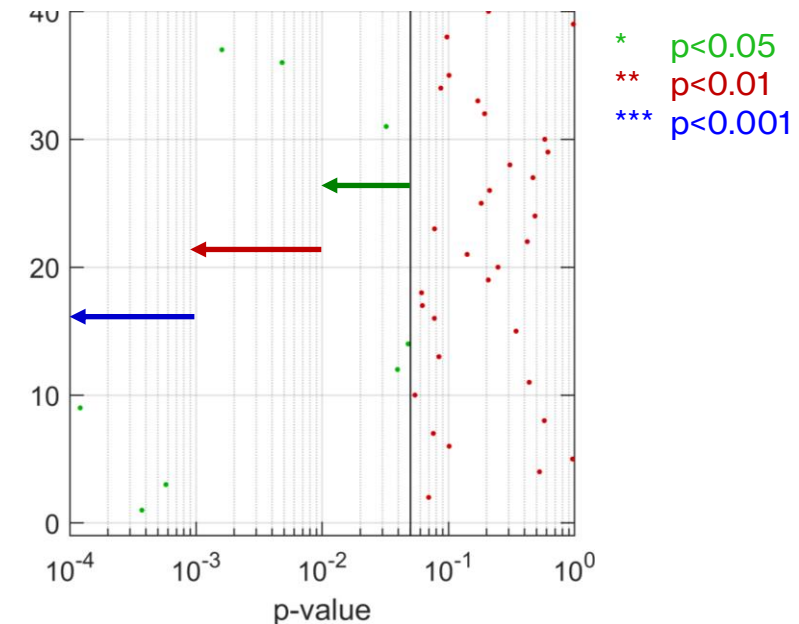


p (roblematic) -values

- Then what on earth are we to do?
- We just have to be careful
 - Interpret a p as a single observation of the random variable P : we can learn from it, but need to be careful
 - If there was no effect, it is less likely that we'd find a small p (but still possible)
 - So a small p does tell us something!
- My suggestion: report confidence interval, along with
 - $p < 0.05$ *
 - $p < 0.01$ **
 - $p < 0.001$ ***
- This allows one to draw some conclusions, without drawing too strong a conclusion



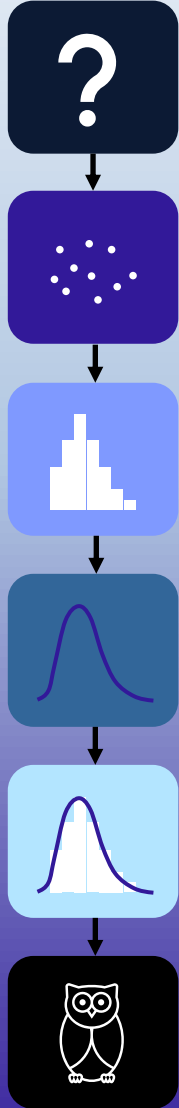
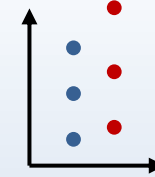
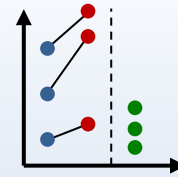
“A frustrated badger”





Summary of z-tests

for reference and understanding



1-sample z-test	Paired 2-sample z-test	Unpaired 2-sample z-test
How does the ...		
population average μ_X	population average difference μ_D	difference between the averages of the two populations, μ_1 and μ_2
... compare to some known value μ_0 ?		
Capture $i = 1:n$ observations, x_i	Capture $i = 1:n$ paired observations, $x_{1,i}$ & $x_{2,i}$ to yield $d_i = x_{2,i} - x_{1,i}$	Capture $i = 1:n_1$ & $j = 1:n_2$ unpaired observations, $x_{1,i}$ & $x_{2,j}$
Visualisation of x_i	Visualisation of $x_{1,i}$ & $x_{2,i}$ and/or d_i	Visualisation of $x_{1,i}$ & $x_{2,j}$
x_i	d_i	$x_{1,i}$ and $x_{2,j}$
... are sampled from (a) normally distributed population(s) with known standard deviation(s) ...		
σ_X	σ_D	σ_1 and σ_2
$\sigma_{\bar{X}} = \sigma_X / \sqrt{n}$ $z = \frac{\bar{x} - \mu_0}{\sigma_{\bar{X}}}$	$\sigma_{\bar{D}} = \sigma_D / \sqrt{n}$ $z = \frac{\bar{d} - \mu_0}{\sigma_{\bar{D}}}$	$\bar{y} = \bar{x}_2 - \bar{x}_1$ $\sigma_{\bar{Y}} = \sqrt{\sigma_1^2/n_1 + \sigma_2^2/n_2}$ $z = \frac{\bar{y} - \mu_0}{\sigma_{\bar{Y}}}$
$H_0: \mu_X$ or μ_D or $\mu_2 - \mu_1$ is not 'different to'/'greater than'/'less than' μ_0 (two-/upper-/lower-tailed) Is observed z likely if H_0 is true, for a given α ? {yes/no}		
We found a mean difference of $\bar{x}$ or $\bar{d}$ or $\bar{y}$ [95% CI] that {was/was not} statistically different from μ_0		

Lecture 6: Introduction to NHST and z-tests

What we covered

- Null hypothesis statistical testing
 - Tailedness
 - NHST outcomes
 - Probability of Type I and Type II errors
 - Drawing proper conclusions from NHST
- Using confidence intervals
- One sample z-test
- Independent two-sample z-test
- Dependent two-sample z-test

Next Up
t-tests and assumptions

